TIRust benchmark

Batched one-loop reduction: TIRust, IREGI, COLLIER and MadLoop

2026-10-25 · AMD EPYC 9754 · CPUs 0 / 254 · identified kernel snapshots

Reading guide. Every displayed timing ratio has TIRust in the numerator: *TIRust / IREGI*, *TIRust / COLLIER*, or *TIRust(candidate) / TIRust(baseline)*. Physical graphs use *TIRust / MadLoop*. Green means TIRust is faster, white parity and red slower. The archival reference-only table shows absolute times and outcomes. Gray is unmeasured; amber marks pending or inconclusive evidence. Source inspection does not establish numerical agreement. Measured TIRust6b24 passes eight numerical cases but retains seven timing regressions/one noisy row. It predates later APIs/preparation. Original-IREGI/COLLIER and physical sessions remain separate; private improvements, including the new unadopted lazy/eager cohort, do not close performance acceptance.

Current priorities

Rank-seven correctness	Complete independently validated IREGI interfaces through rank seven	**Pending**
Higher-rank performance	Profile the fastest path after correctness; retain all measured snapshots	**Next**
Complete evaluation	Six physical rows measured; two failures retained	**Inconclusive**

1 Current coverage

The local Python interface and example gallery execute successfully with repaired wheel004. Complete mathematical acceptance remains open. Low-rank COLLIER speed is not a completion gate under the latest user priority; all historical measurements below retain their original source snapshots.

Interface / domain	Executed scope or remaining gate	Status
Local PyO3 wheel	17/17 numerical and 12/12 interoperability tests, including the actual SVG renderer; installed consumer typing has zero errors/warnings.	Passed
Dotted propagators	Generalized powers/dimension shifts have their strongest retained coverage at N≤4. The current wheel/gallery checks the dotted bubble against its mass Dual derivative.	Scoped
Example gallery	4/4 native test methods; three successful HTML exports; all three rendered at desktop and narrow dark sizes in real Chromium.	Passed
Tensor coverage	Actual Cargo N1–N5 rank7 massive arrays, massless N2–N4 arrays and N6 ranks4–7 pass retained-family checks at Float128/256. N6 lifecycle/contractions pass. Public N7 is still scalar-only.	Scoped
Performance	Historical private selected TIRust/COLLIER: seven slower families and one short inconclusive family. The fresh scheduler cohort is ineligible because of host contention; no default adoption.	Measured

Local Linux/Python 3.13 execution qualifies the named examples and focused API tests. Release distributions, other platforms, separate Community-host loading and a reactive numerical browser frontend remain unqualified. C0/D0 mass-Taylor identities now pass on wheel004 at working 128/256 bits with requested accuracy 53. Historical broader mixed-Dual finite-difference evidence remains separate; these identities do not rerun that complete suite.

2 Rank-seven correctness checkpoint

Priority: finish validated IREGI through rank seven. Profile whichever implementation is fastest, concentrating subsequent optimization on higher ranks. Low-rank speed against COLLIER is not a completion gate. The complete objective remains open; historical timings retain their measured contexts.

Domain	Executed scope	Outcome
N1–N5 / rank7	Actual Cargo PV: three massive samples per topology, Float128/256, 39,600 complex reference coefficients and 19,800 precision pairs.	Pass[1]
Massless N2–N4	Complete rank7 arrays: 7,920 reference comparisons and 3,960 precision pairs. Separate UV, IR and IR² slots retained, including the scaleless bubble.	Pass[1]
N6 / rank7 batches	21,120 tensor and 128 sparse/dense contraction coefficients; empty, mixed failed lanes, reorder/chunks, cloned scratch and malformed-call recovery.	Pass[1]
N6 / ranks4–7	Separately requested plans, Float128/256: 5,888 reference coefficients and 2,944 precision pairs. All previous rank7 prefixes retained.	Pass[1]
N7 / rank0	Actual Cargo Float128/256 singleton agreement. Nine-call lifecycle stopped after six completed calls; later scratch/recovery checks remain open.	Partial
N6 / rank7 COLI	Native accuracy flag −2; 315 missing diagnostic coefficients and one finite discrepancy. All three pole probes return failure sentinels.	Rejected
Full rank-seven goal	Public N7 tensors, broader methods/kinematics/numeric types, generalized tensors, high-point Dual and frontend integration still need qualification.	Open

[1] Retained massive and massless families, native raw tensor components, requested accuracy 53 and working precision 128/256. Every moment includes all four native Laurent slots. Independent-reference gates remain 10^{-10}; precision and scaling gates remain 2^{-80}. Maximum reference discrepancies are 1.839×10^{-16} for massive N1–N5, 2.522×10^{-16} for massless N2–N4 and 1.265×10^{-16} at N6. Working precision does not assert full working-precision accuracy. These cohorts retain their exact captured source contexts, including the older N6 lifecycle context before the Cayley change. No kernel timing claim follows.

2.1 Complete arrays, batches and preparation

The N1–N5 corpus contains 30 successful sample lanes and passes 13,200 uniform scaling checks, 19,800 exact IR-zero checks and 996 convergent UV-zero checks. Original IBP supplies the reference arrays. Three original bubble PV requests produce no result; six stable triangle/box PV arrays disagree in 486 finite positions. Those failures remain visible and are never substituted as accuracy anchors.

The N6 rank-seven batch/contraction cohort passes at both precisions with 16 successful and eight explicitly rejected lanes. Sparse odd/mixed support and dense coefficients covering all 330 moments use the existing Symbolica contraction evaluator. The maximum contracted-reference discrepancy is 5.090×10^{-18}. All actual planar-child fallbacks are retained. Cloned scratch was exercised serially; concurrent execution is a separate gate.

The adopted routing and eligible five-point Cayley preparation uses Symbolica rational-polynomial matrices and exact identity certificates. Other syntax retains AtomField. Physical singularities preserve exceptional recurrence; conversion/certificate errors remain preparation failures. All 51 focused IBP tests pass. The subsequent literal N6 rank4–7 checks cover complete arrays in this source context. No local algebra solver or numerical specialization was introduced.

Typed-Dual malformed-input failures now clear earlier counters, profiling times and diagnostic messages. Two public regression tests pass, including a genuine mass derivative and valid recovery. Initial test-driver failures remain saved. This diagnostic fix does not change numerical formulas or qualify high-point Dual inputs.

2.2 Reference accuracy adjudicated

Original IREGI and its bundled OneLOop were built with uniform quad arithmetic, retaining the original complex-matrix sources, formulas, thresholds and limited-digit constants. Runtime type canaries and four scalar normalization controls pass. Stable IBP agrees with both complete TIRust arrays. The older binary64 IBP comparison still fails 90 slots; those failures remain preserved. Quad PV is unstable and disagrees in 139 positions. Quad Hybrid reports stability but disagrees in 19, and remains unqualified. Stored 113-bit arithmetic does not establish achieved 113-bit result accuracy.

The rejected COLI capture is not an independent accuracy witness. Its zero explicit Laurent-pole residual count does not turn its failed pole-parameter probes into passing checks. No high-rank kernel timings are inferred from validation-stage durations.

Saved-evidence replay, without new integral evaluation: `python3 validation/public-rank7-current-cargo-n1-n5/check.py`
`python3 validation/highpoint-six-point-rank7-lifecycle/check.py`
`python3 validation/highpoint-six-point-literal-ranks4-7-current-cargo/check.py`
`python3 validation/quad-original-iregi-small-n-rank7/check.py`. The separate rejection record is `validation/collier-n6-rank7-diagnostic/qualification.json`.

2.3 Massless poles and shared execution

The three massless points separately exercise a lightlike scaleless bubble, an on-shell triangle and a scattering box. Original quad IBP supplies all three reference arrays. All six Float128/256 arrays pass; the largest normalized reference discrepancy is 2.522×10^{-16}. The largest precision-pair discrepancy is 5.891×10^{-37}. All six actual routes are analytic PV, with no fallback. In particular, the bubble's nonzero UV and IR single poles are retained separately even though their sum vanishes. No positive-mass IR-zero rule or zero projection is applied to these data.

This cohort uses the same frozen actual-Cargo artifact as the earlier N1–N5 arrays. It does not qualify later source edits or the whole massless domain. The independent reviewer reconstructs all 11,884 comparison records. Per-case terminal records pass; controller phase order and maximum concurrency are executor-attested because absolute UTC start/finish records were not saved. Elapsed durations are diagnostics, not performance measurements.

The additive `TensorReductionPlan` / `TensorWorkspace` facade shares eligibility and contraction with the existing generic, six-point and seven-point engines. Six ordinary tests and a separately selected actual Float128 scalar delegation test pass. The latter checks N6/N7 direct-wrapper equality, invalid-lane isolation, contractions, nested execution counters and empty-call reset. It introduces no recurrence or scalar formula. Explicit Dual layouts, profiling and nondefault policies retain generic execution; requested algorithms and compiled scalar-master defaults are preserved. These core tests alone do not qualify new CLI or Python wheel builds.

Separate rank-seven IBP captures complete at N1/N3 and pass 15,840 reference comparisons, 7,920 precision pairs and 5,280 scaling identities. Both N2 captures time out during workspace preparation and produce no coefficient array. The corpus remains incomplete. Original cleanup failures and later process-absence observations are retained separately. No timing gain or complete-method qualification follows from the available results.

The private seven-point probe completes ranks one and two at one regular positive-mass point, passing 160 complex reference comparisons and 80 precision pairs. Its earlier rank-seven attempts produce no arrays: Float128 reaches its preparation deadline, and Float256 is deliberately stopped before its deadline after the same preparation/resource problem. Both original cleanup failures remain recorded; later process-absence observations are additive. These are private positive-rank witnesses and failed rank-seven attempts, not public rank-seven qualification.

A targeted Symbolica compiler candidate reuses an exact SSA condition's zero/nonzero fact inside its dominating branch. The numeric condition value is preserved, and branch facts are restored before siblings and function scopes. Seven new and eleven relevant existing tests pass, including mixed higher Duals and instruction-major batching. The initial optimized test build times out; a test-only opt0/debug0 build completes with the same sources and features. These engineering records establish focused compiler semantics; the numerical N2/N7 reruns remain separate gates.

Additional saved-evidence replay: `python3 validation/public-rank7-massless-ir/check.py`
`python3 validation/shared-tensor-facade/check.py`
`python3 validation/public-rank7-ibp-owned-artifact-n1-n3/check.py`.

3 Additional correctness milestones

These retained private results support the remaining implementation work. They do not expand the qualified public domain by themselves.

Scope	Complete saved checks	Status
Generalized N6/N7	N6 dotted, N7 dotted and N7 h=−1: 240 complex outputs at Float128/256; original agreement, scaling, repeated/chunked batch profiles and 80 exactly retained control values.	Private pass
N6 scalar Dual	19 tests and four captures: 528 parent Taylor slots, 156 packed child rows and 264 precision pairs. Independent first mass derivative/dotted anchor and coherent dilation jets.	Private pass
N7 scalar API	35 focused actual-Cargo tests; singleton Float128/256 checks pass. The partial lifecycle retains 28 moment and 28 unit-contraction slots, including rejected neighbors and reordered/chunked inputs.	Partial

The generalized preparation change uses Symbolica's existing rational-polynomial matrix operations, with exact reconstruction certificates and an AtomField fallback. It also distinguishes a singular matrix from a preparation failure. The retained cohort passes 7,415 independent saved-data checks, covering all 2,060 input slots and all complete outputs. The preparation adapter is now adopted into the main Cargo source and passes the ordinary-array checks above. This older generalized numerical cohort retains its private dependency context; it has not been rerun in the new Cargo context. Higher-rank generalized requests, genuine D6 domains, complex masses and Dual directions remain separate work.

The component-sensitivity prototype transports complete heuristic Taylor sensitivity operands rather than padding derivative entries with zeros. Its mass derivative agrees with independent dotted IBP/original anchors; its coherent scale jets pass homogeneity and precision checks. These are numerical sensitivity estimates, not rigorous derivative uncertainty enclosures. General independent mixed directions, finite differences and public N6/N7 Dual constructors remain unqualified. The rectangular mixed-direction follow-up is source-only.

The N7 lifecycle stopped at its fixed deadline during the independent-scratch call. Six preceding calls are retained; the remaining scratch, later empty and malformed-call recovery checks are not silently treated as passing. The first failed Float256 attempt also remains alongside its separately bounded successful follow-up. Positive-rank seven-point validation is ongoing.

Exact-master preparation-cache experiments are preserved separately: their 38-request whole-output regression passes, while earlier typed schedules remain incomplete. No cache candidate or timing gain is adopted by this checkpoint. No new high-rank benchmark measurements are added here.

Sources and saved-evidence replay: `validation/routing-rational-cayley-full-core/` and `validation/highpoint-dual-sensitivity/`. The actual N7 source/build boundary is recorded in `validation/highpoint-seven-point-current-cargo/qualification.json`.

4 Historical kernel baseline: `4c32ece`

Table 1: **Massive bubble/triangle, binary64, hybrid mode.** TIRust and IREGI total medians in µs/point. **TIRust / IREGI** ratios are paired-block medians; intervals are 99% bootstrap confidence intervals. Batch 1 is latency; batch 32 is amortized cost per point. Each backend uses its own paired reference.

Case	Rank	Interpreter Batch 1	Interpreter Batch 32	Native Batch 1	Native Batch 32
Bubble	0	3.176 µs IREGI 1.988 µs · ×1.598 99% CI 1.585–1.607	1.708 µs IREGI 1.885 µs · ×0.906 99% CI 0.902–0.911	0.936 µs IREGI 0.943 µs · ×0.993 99% CI 0.987–1.003	0.903 µs IREGI 0.878 µs · ×1.031 99% CI 1.024–1.042
Bubble	1	9.552 µs IREGI 5.707 µs · ×1.670 99% CI 1.655–1.688	5.831 µs IREGI 5.748 µs · ×1.015 99% CI 1.008–1.019	4.663 µs IREGI 3.044 µs · ×1.534 99% CI 1.526–1.549	3.231 µs IREGI 3.029 µs · ×1.069 99% CI 1.062–1.081
Triangle	0	29.152 µs IREGI 8.496 µs · ×3.446 99% CI 3.415–3.454	18.229 µs IREGI 8.433 µs · ×2.160 99% CI 2.147–2.172	4.227 µs IREGI 4.968 µs · ×0.849 99% CI 0.844–0.858	8.405 µs IREGI 4.899 µs · ×1.715 99% CI 1.704–1.723
Triangle	1	38.505 µs IREGI 18.297 µs · ×2.101 99% CI 2.088–2.108	25.634 µs IREGI 18.224 µs · ×1.408 99% CI 1.400–1.413	8.809 µs IREGI 10.418 µs · ×0.850 99% CI 0.844–0.854	12.461 µs IREGI 10.213 µs · ×1.220 99% CI 1.206–1.228



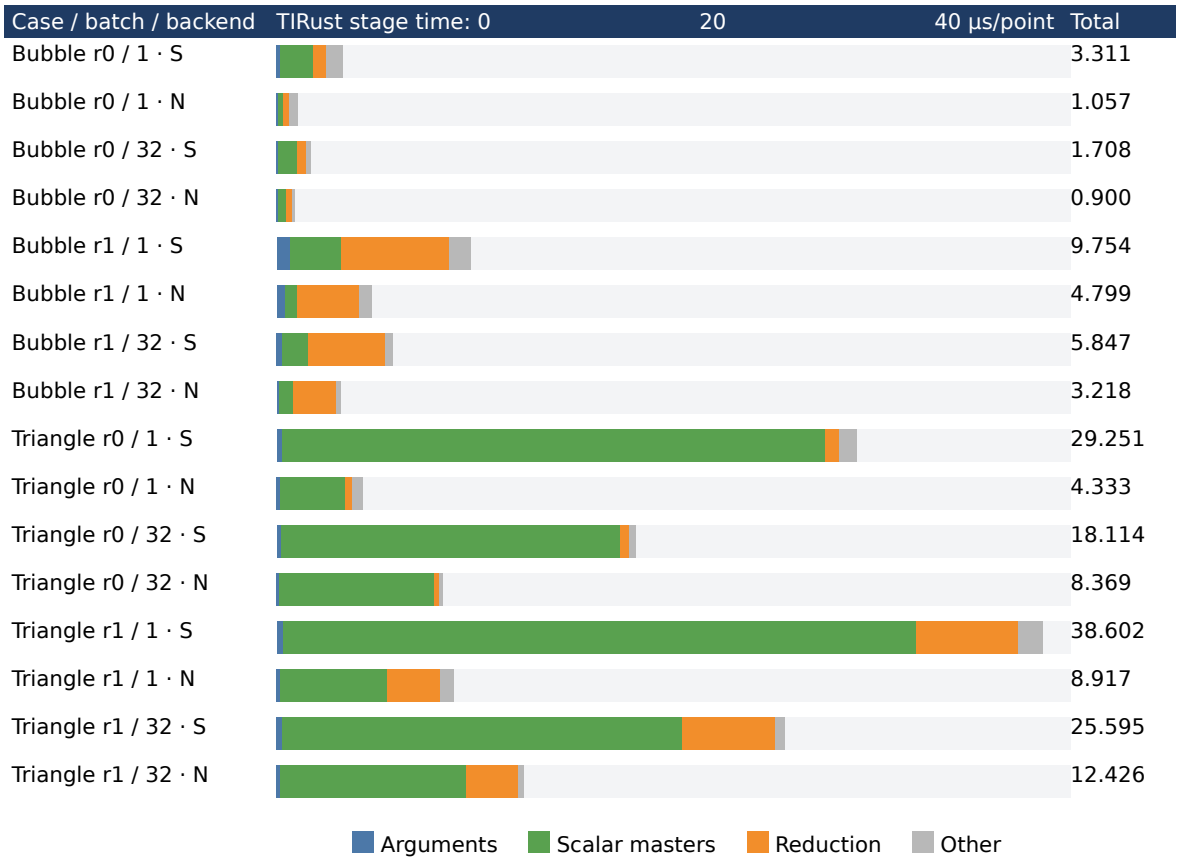
Both backends pass all eight numerical checks at 10^{-10}. The interpreter has seven regressions and one improvement; native has five regressions, one parity result and two improvements. **Absolute IREGI timings changed substantially between sessions**, despite identical inputs/reference binary and CPU settings. Each verdict uses its own paired run; no direct backend speedup is inferred across sessions. Both use the historical `4c32ece` revision; higher-rank performance remains unqualified.

The newer optimized interpreter snapshot `5c9e3dd` passes eight numerical checks, with six timing regressions and two noisy rows. The `80949db` snapshot gives five regressions/three noisy rows. Latest `6b24c03` passes all eight checks: seven regressions/one noisy row. Separate paired comparisons follow; performance acceptance is unmet.

Newer validation includes four integrated graphs. The top pentagon at Float 128/256 agrees with PV/hybrid and MadLoop; original IBP still fails. The latest physical timing comparisons remain inconclusive.

5 Historical stage profiles: 4c32ece

TIRust medians from separate instrumented passes, in µs/point. All bars use 0–40 µs/point; S denotes the Symbolica interpreter, N the generated native backend. Sessions are separate: compare each with its own reference in the preceding baseline table, not with the other backend's absolute bar length. Raw ns/batch are divided by 1000 × batch width.



"Other" is total median minus named stage medians, not a separately timed region. Scalar-master time includes complete master calls; reduction includes stability checks. IREGI has no measured stage split here. For native triangle rank 0, scalar masters cost 3.282 µs at batch 1, but 249.977 µs per batch 32, or 7.812 µs/point: this batch path has higher amortized master cost. The observation identifies a profiling target, not a cross-session speedup or a proven cause. Full stage values and samples remain in the raw data.

5.1 Preparation and measurement conditions

Backend	Bubble r0	Bubble r1	Triangle r0	Triangle r1
Interpreter	1.572 s	0.143 s	0.104 s	0.091 s
Native	0.011 s	0.012 s	0.001 s	0.013 s

Preparation values are single observations without repeated uncertainty, excluded from warm kernel ratios. Batch32 reuses the plan; numerator preparation is outside this tensor-array benchmark.

Both backends of the isolated pushed revision `4c32ece` ran on AMD EPYC 9754 CPU 0, with 5 warm-ups and 21 paired blocks targeting 30 ms, randomized order with seed 271828. TIRust used release opt-level3, thin LTO and one codegen unit; Fortran used `-O3`. Structural plans and allocation capacity were reused; numerical results were recomputed, and original recurrence caches were freed per sample. Subprocess transport is excluded. All source/binary provenance checks pass, and every ratio interval is no wider than 2% of its median. Raw samples, inputs, flags and hashes are retained in `validation/benchmark/baseline-4c32ece/` and `validation/benchmark/native-4c32ece/`.

A separate frozen-native policy experiment keeps `Whole`: all 18 C0/D0 comparisons across three orderings find `FixedRows4/8/16` slower, with per-comparison 99% intervals excluding parity. It uses 32 valid lanes, CPU 0 and 21 rotated blocks. This selects a batch policy, not IREGI speedup; see `validation/native-policy/`.

Native allocator adoption remains rejected. Historical 24 timing rows give two singleton parities/22 noisy rows; new D0 timings include two batch32 regressions. D0 exact old/new parity passes 40 configurations/1,488 complex outputs (f64/DD/Float256); interpreter differences remain. Separate candidate decisions appear in the optimization ledger and `validation/native-stack-batch/`.

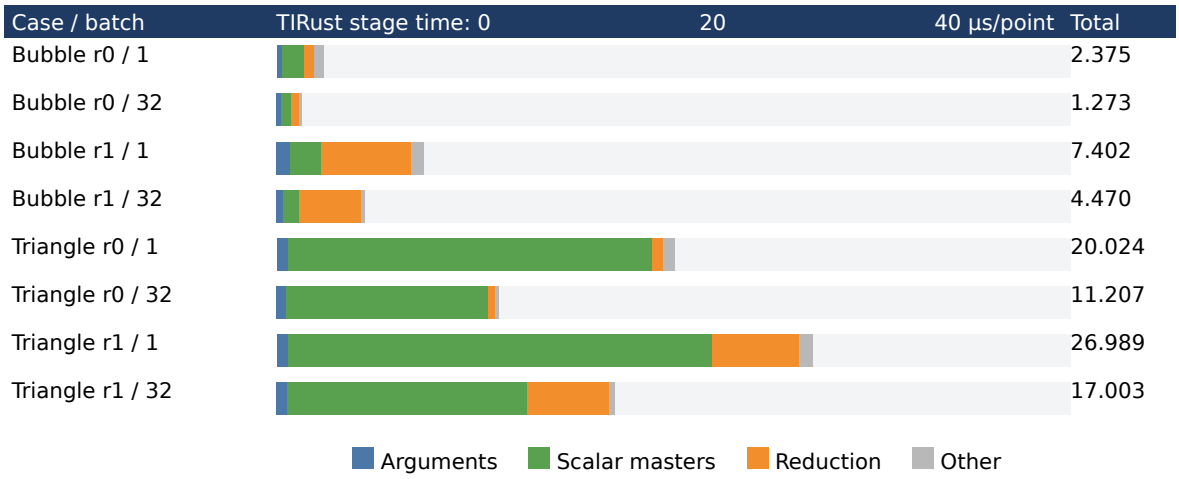
6 Latest kernel snapshot: `6b24c03`

All eight massive bubble/triangle checks pass 10^{-10}, with maximum normalized error 2.46×10^{-13}. Seven timing rows regress; the rank-one triangle singleton exceeds the unchanged 2% interval-width limit (2.357%) and remains **inconclusive**. This measured snapshot predates the standalone planar and finite-D0 certification APIs and later expression preparation. The performance requirement remains unmet.

Table 2: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.225 µs IREGI 0.948 µs · ×2.344 99% CI 2.335–2.353 Regression	1.265 µs IREGI 0.896 µs · ×1.413 99% CI 1.407–1.416 Regression
Bubble	1	7.245 µs IREGI 3.048 µs · ×2.388 99% CI 2.372–2.403 Regression	4.498 µs IREGI 3.046 µs · ×1.480 99% CI 1.467–1.488 Regression
Triangle	0	19.618 µs IREGI 4.925 µs · ×3.986 99% CI 3.977–3.996 Regression	11.152 µs IREGI 4.963 µs · ×2.246 99% CI 2.237–2.255 Regression
Triangle	1	26.777 µs IREGI 10.356 µs · ×2.586 99% CI 2.570–2.631 Inconclusive: noise	16.967 µs IREGI 10.411 µs · ×1.627 99% CI 1.621–1.636 Regression

6.1 Separately instrumented component timings



Stages use a separate instrumented pass. "Other" is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 254, five warm-ups and 41 randomized paired blocks targeting 100 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`.

The complete optimized workspace build exits successfully. All 414 immutable build-source hashes, both binaries, feature selection and harness match the retained provenance. Other project computations were held. CPU 254 was selected by a read-only load survey because unrelated work occupied CPU 0; external host activity remained uncontrolled. No system settings were changed. Historical sessions used different CPUs and timing protocols; only each session's paired reference supports its verdict, with no causal revision speedup. Raw samples and reproduction: `validation/benchmark/baseline-6b24c03-licensed/`.

7 Direct TIRust / IREGI / COLLIER comparison

Private optimized prototypes compare the complete reduction call on the same massive Euclidean N2/N3 rank0/1 inputs. **B** is baseline TIRust with Symbolica masters; **S** is the same TIRust prototype selecting native A0/B0/C0 for plain f64. IREGI uses Hybrid and COLLIER uses COLI mode 1. Every ratio below has TIRust in the numerator; all forty results remain visible.

Table 3: **Same-cohort paired latency ratios and 99% CIs.** B and S denote TIRust variants, not separate reference libraries. Below one favors the numerator. Amber preserves the short-block family.

Family / batch	TIRust B / IREGI	TIRust S / IREGI	TIRust B / COLLIER	TIRust S / COLLIER	TIRust S / TIRust B
Bubble r0 batch 1	2.381 2.374–2.386 Short block	1.550 1.546–1.554 Short block	2.949 2.937–2.962 Short block	1.921 1.914–1.929 Short block	0.651 0.650–0.652 Short block
Bubble r0 batch 32	1.539 1.535–1.542 Slower	1.383 1.379–1.385 Slower	1.777 1.769–1.782 Slower	1.594 1.587–1.602 Slower	0.899 0.897–0.900 Gain
Bubble r1 batch 1	2.347 2.332–2.355 Slower	2.023 2.013–2.034 Slower	6.100 6.078–6.132 Slower	5.274 5.258–5.289 Slower	0.864 0.861–0.869 Gain
Bubble r1 batch 32	1.496 1.487–1.503 Slower	1.459 1.452–1.466 Slower	3.920 3.911–3.927 Slower	3.818 3.812–3.823 Slower	0.974 0.972–0.977 Gain
Triangle r0 batch 1	4.071 4.062–4.076 Slower	0.985 0.983–0.987 Gain	5.311 5.301–5.316 Slower	1.286 1.284–1.288 Slower	0.242 0.242–0.243 Gain
Triangle r0 batch 32	2.402 2.394–2.407 Slower	1.870 1.867–1.875 Slower	3.103 3.096–3.109 Slower	2.417 2.413–2.423 Slower	0.779 0.778–0.781 Gain
Triangle r1 batch 1	2.651 2.644–2.655 Slower	1.057 1.054–1.059 Slower	4.484 4.480–4.494 Slower	1.790 1.787–1.791 Slower	0.399 0.398–0.399 Gain
Triangle r1 batch 32	1.708 1.701–1.718 Slower	1.414 1.406–1.423 Slower	2.885 2.877–2.898 Slower	2.388 2.384–2.400 Slower	0.828 0.825–0.832 Gain

Selected versus COLLIER: seven regressions, one inconclusive family. Selected versus IREGI has six regressions, one gain (triangle rank 0 singleton) and one inconclusive family. Selected versus baseline has seven gains and one inconclusive family. Baseline/reference comparisons each retain seven regressions and one inconclusive family. No production/default adoption or complete performance acceptance follows.

Explicit posthoc analysis revision2. The original frozen analyzer marked five ratios short-block-inconclusive and 35 cohort-failed because one local short block set the aggregate status. A separately reviewed saved-data correction verifies clean termination and every original global check before assigning family gates. All forty sample arrays, medians, CIs, widths and numerical order-stratum statistics are unchanged; the conservative verdict remains archived. No new timing, threshold change, omitted family or repeated cohort.

One COLLIER block in bubble rank 0/batch 1 lasts 99.893971 ms, below 100 ms; all five ratios of that family remain inconclusive. Protocol: CPU 254, ten warm calls, 48 quartets per family containing all 24 permutations of the four engines twice, shared fixed iterations, 100 ms blocks, 10,000 paired bootstrap resamples, 99% CIs and 2% width gate. Every complete tensor/Laurent array and native pole probe passes before and after timing. Transport is excluded. This is a source-bound historical prototype, not the current production build. Sources: `validation/selected-native-four-engine/v3-frozen/` and `validation/selected-native-four-engine/posthoc-revision2/`.

8 Archival reference timings: IREGI and COLLIER

Identical massive Euclidean tensor integrals compare original IREGI Hybrid (mode 2) with COLLIER 1.2.10 COLI (mode 1). Each row requests the complete Cartesian tensor array and all four native Laurent slots. This is a reference-library comparison, separate from the TIRust performance gate. No TIRust/COLLIER ratio is inferred from different sessions. The direct four-engine cohort has its own separate source-bound table.

Table 4: **Archival binary64 reference timings, µs/point.** Batch 1 is latency; batch 32 is amortized cost per point. Qualified comparison outcomes are retained without reference/reference ratio displays.

Topology / rank	Batch	IREGI Hybrid	COLLIER COLI	Qualified outcome
Bubble rank 0	1	0.943	0.764	Inconclusive: short block
Bubble rank 0	32	0.927	0.766	COLLIER faster
Bubble rank 1	1	3.129	1.173	COLLIER faster
Bubble rank 1	32	3.122	1.172	COLLIER faster
Triangle rank 0	1	4.994	3.781	COLLIER faster
Triangle rank 0	32	4.963	3.825	COLLIER faster
Triangle rank 1	1	10.752	6.148	COLLIER faster
Triangle rank 1	32	10.864	6.145	COLLIER faster

The 99% interval width must be at most 2% of the paired median, and every block must reach 100 ms. Amber rows remain inconclusive. Raw ns/batch values are divided by 1000 × batch width for this display. Original paired ratios and intervals remain archival metadata in the JSON.

Numerical scope. All eight families pass the unchanged dimensionally normalized 1e-10 criterion; maximum error is 2.49e-13. The bubble rank0 singleton includes a 98.438 ms block below the 100 ms requirement and remains inconclusive; no row was rerun or omitted.

Cache policy. within-sample recurrence memoization in both; no cross-sample/event numeric reuse

COLLIER timing includes invariant setup and native conversion; IREGI includes mass setup and cache disposal. Both include the full tensor call and status checks.

Measurement protocol. CPU 254, five warm-ups and 41 balanced randomized paired blocks; common fixed iteration counts, registered minimum 100 ms. Both builds use GNU Fortran 15.2.0 -O3 without fast-math. Project computations were held; unrelated host activity remained uncontrolled. Transport and regulator probes are outside timing.

Ordinary non-dotted, fixed-dimension cases have zero IR poles; regulator probes are untimed. This archival timing cohort does not qualify higher points/ranks, singular or IR-divergent families or physical numerators. Raw observations and reproduction: `validation/collier-reference/timing-initial8/summary.json`.

9 Extended reference coverage: N2–N7

A separate correctness-only capture extends the ordinary, massive Euclidean reference matrix. Each cell keeps the native COLLIER admission status; numerical output from a rejected call cannot qualify a comparison. This table contains no additional timings and no new TIRust performance claim.

Table 5: **All 34 reference rows, including failures.** COLI-admitted means native status accepted and the complete native tensor/Laurent array agrees with IREGI within 10^{-10}. This table does not establish independent higher-point identities.

Propagators / rank	Batch 1	Batch 32
N2 / rank 0	COLI admitted	COLI admitted
N2 / rank 1	COLI admitted	COLI admitted
N3 / rank 0	COLI admitted	COLI admitted
N3 / rank 1	COLI admitted	COLI admitted
N4 / rank 0	COLI admitted	COLI admitted
N4 / rank 1	Unqualified	Unqualified
N4 / rank 2	Unqualified	Unqualified
N4 / rank 3	Unqualified	Unqualified
N5 / rank 0	COLI admitted	COLI admitted
N5 / rank 1	COLI admitted	COLI admitted
N5 / rank 2	COLI admitted	COLI admitted
N6 / rank 0	COLI admitted	Unqualified
N6 / rank 1	Unqualified	Unqualified
N6 / rank 2	Unqualified	Unqualified
N7 / rank 0	Unqualified	Unqualified
N7 / rank 1	Unqualified	Unqualified
N7 / rank 2	Unqualified	Unqualified

17 COLI-admitted, 17 unqualified rows. The six N5 rank 0–2 rows have maximum normalized disagreement 4.76×10^{-12}; the N6 rank 0 singleton reaches 2.60×10^{-13}. Existing N4 rank 1–3 rejections remain visible. Other higher-point calls retain COLI accuracy flags −1/−2 and DD error flag −10. Both policies and all raw statuses remain in the evidence; library code and rejection criteria are unchanged.

Attempt004 retained 239 validated isolated responses. Attempt005 appends only its eight missing requests, with no repeated capture or timing. The final N7 rank 2/batch 32 case retains unstable original PV, a nominally successful IBP result disagreeing with Hybrid by 0.520 normalized, and rejected COLLIER outputs. The N6 rank 2 mass-deformed IREGI disagreement of 1.97 also remains unresolved.

The independent saved-artifact/lane-gather audit verifies 81,086 assertions, 54,224 admitted native complex pairs and 5,427 rejected null slots. This is distinct from independent mathematical identities. IR-divergent, dotted and shifted-dimension COLLIER coverage is unqualified. Sources: `validation/collier-reference/shared/attempt004/` and `validation/collier-reference/shared/attempt005/`.

10 Independent checks and retained failures

10.1 Massive N5 reference identities

Three retained positive-F massive, undotted N5 branches at ranks 0–2 pass independent prefix, pinch/denominator/trace and cyclic/reversed-routing checks. Batch 32 cycles those same three branches; it does not add 32 distinct kinematic configurations. These synthetic checks use the reference codes, separately from TIRust reduction, physical numerators and MadLoop.

Independent check	Residual records	Complex slots	Maximum normalized error
Rank prefixes	924	3696	0
Pinch / denominator / trace	756	3024	3.135×10^{-12}
Cyclic / reversed routing	360	1440	4.101×10^{-12}
Finiteness / pole zeros	3357	10071	0

All 5397 residual records / 18,231 complex comparisons pass the unchanged 10^{-10} gate. The saved-data audit independently verifies 511,509 assertions and 346 archive members, including 465 admitted native lanes and 13,428 unique native complex pairs. The earlier capture's audit-pending status is immutable; the separate completed audit supplement supplies this result. No higher-rank, singular, dotted, dimension-shift, physical or timing qualification follows. Source: `validation/collier-reference/shared/identities006/correctness001/audit001/`.

10.2 Private N7: exact tests only

A frozen private N7 source build passes 17 exact tests. Its ignored generic variant-constructor test remains unrun. Source/build/test receipt review is distinct from numerical integral evaluation: no N7 full-array agreement, precision convergence, public adoption or performance is established. Source: `validation/highpoint-thin-n7-exact/`.

10.3 Private N6 rank 7: failed Float128 comparison

The retained Float128/requested-53 singleton returns 330 moments / 1320 native complex Laurent slots, including 990 exact zero pole slots. Its unchanged original-IBP comparison fails 90 finite slots: 12 at rank 6 and 78 at rank 7. Maximum normalized error is 2.539×10^{-9} against 10^{-10}. A completed process and reported stable status do not qualify the complete array.

For this real positive-F point, the independent analytic real-zero identity has no TIRust violations; original IBP and Hybrid each violate it in 88 slots. This explains 88 of the failed original-IBP complex comparisons from the reference real residual alone. It does not certify the remaining imaginary values or overturn the failed complete-array gate. Original PV stays explicitly unstable. Two saved quadrature witnesses are limited diagnostics, without rigorous error bounds or a causal cache-bug conclusion.

A separate Float256 diagnostic saves the same 1320 slots and 990 exact zero poles. Across Float128/256, the maximum observed normalized difference is 5.421×10^{-32}, below 2^{-80}. Original IBP, PV and Hybrid still disagree in 90, 151 and 109 slots respectively; unstable PV remains unqualified. This observed refinement is diagnostic: complete-array accuracy and precision-convergence acceptance remain false, with no high-rank adoption, lifecycle or timing result. The immutable saved-data packet and its initial missing-file failure are preserved at `validation/checkpoint-audits/2026-10-24-thin-n6-rank7-128/`. The separate terminal capture and independent audit are at `validation/highpoint-thin-n6-rank7-phase/`.

One-component quadrature diagnostic. For moment 126, q_0^6, six fixed-rule calculations (orders 12/16/20, two maps) give $(0, -0.05166894942216114)$. The estimated normalized uncertainty is 3.113×10^{-15} and is nonrigorous. Discrepancy plus that estimated margin is 3.249×10^{-15} for TIRust Float128, within the 10^{-10} diagnostic gate; original IBP and Hybrid give 4.274×10^{-10} and 4.267×10^{-10}, outside it. PV remains natively rejected. This supports one finite component; complete-array acceptance stays failed. Source: `validation/temporal-sixpoint-oracle/q0-sixth-001/qualification.json`.

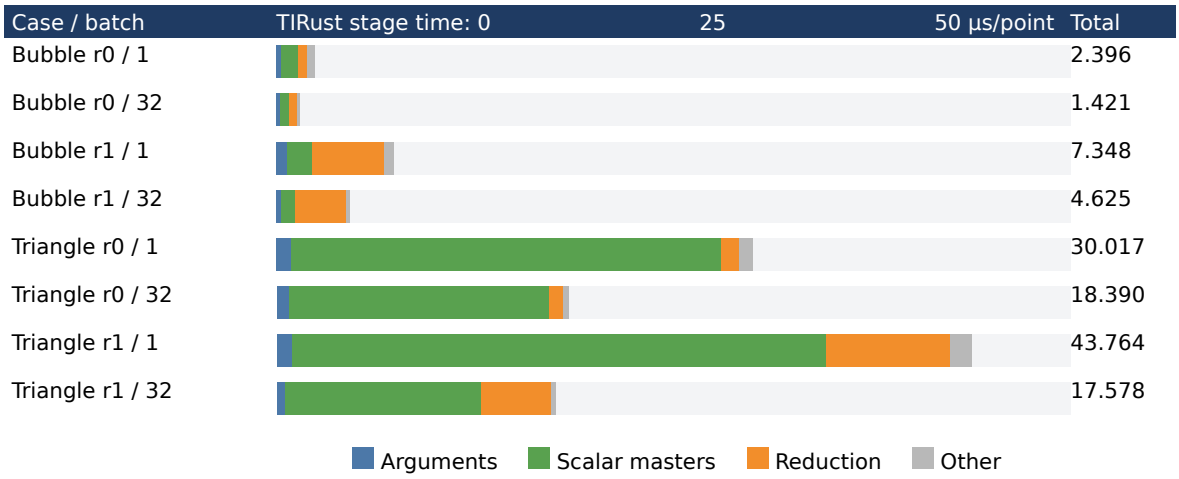
11 Historical kernel snapshot: `80949db`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion, maximum normalized error 2.46×10^{-13}. Five timing rows regress; three exceed the unchanged 2% interval-width limit and remain **inconclusive**. This optimized snapshot includes the adopted master stack allocation, generalized PV and lazy Cayley solve. Later core-allocation and experimental changes are excluded. The performance requirement remains unmet.

Table 6: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.278 µs IREGI 0.946 µs · ×2.403 99% CI 2.389–2.420 Regression	1.380 µs IREGI 0.892 µs · ×1.550 99% CI 1.537–1.557 Regression
Bubble	1	7.243 µs IREGI 3.083 µs · ×2.342 99% CI 2.318–2.361 Regression	4.619 µs IREGI 3.077 µs · ×1.500 99% CI 1.490–1.518 Regression
Triangle	0	19.553 µs IREGI 4.967 µs · ×3.927 99% CI 3.911–3.944 Regression	17.604 µs IREGI 7.296 µs · ×2.374 99% CI 2.325–2.432 Inconclusive: noise
Triangle	1	43.420 µs IREGI 17.458 µs · ×2.500 99% CI 2.463–2.578 Inconclusive: noise	28.540 µs IREGI 17.009 µs · ×1.672 99% CI 1.490–1.762 Inconclusive: noise

11.1 Separately instrumented component timings



Stages use a separate instrumented pass. “Other” is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`.

All 401 immutable build-source hashes, both binaries and the harness match the retained provenance. Other project computations were held; a pending native compiler was suspended. Host activity outside the project was not controlled. The accepted license contains no tracked key material. This is a separate session from the historical tables; its absolute timings establish no revision speedup. Raw samples and reproduction commands: `validation/benchmark/baseline-80949db-licensed/`.

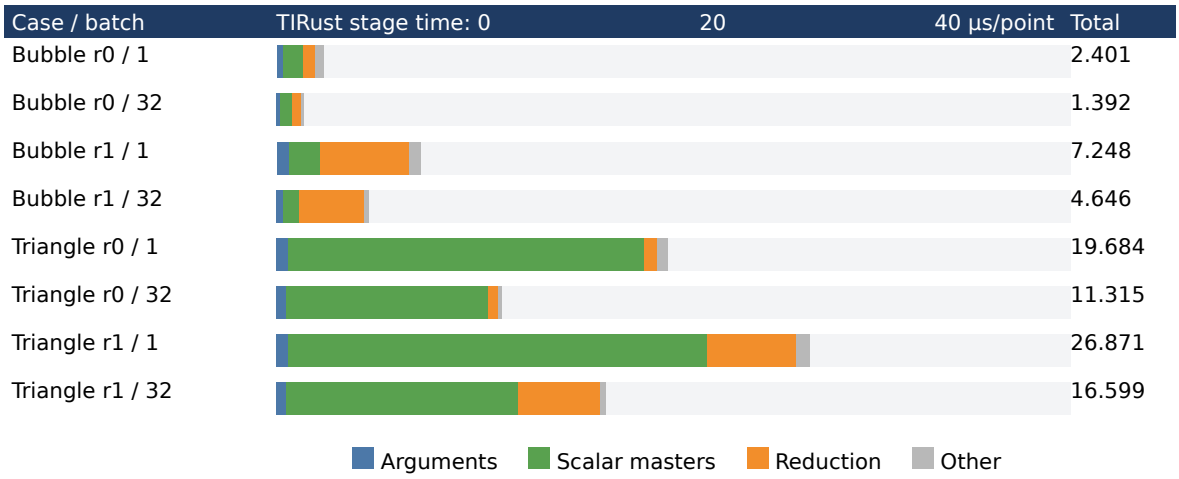
12 Historical kernel snapshot: `5c9e3dd`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion. Six timing rows regress; two exceed the allowed 2% relative confidence-interval width and remain **inconclusive**. Amber cells denote noise, not improvement. This commit predates default stack reuse, generalized PV and the vector-solve candidate. The overall performance requirement is unmet.

Table 7: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.271 µs IREGI 0.955 µs · ×2.381 99% CI 2.364–2.393 Regression	1.389 µs IREGI 0.901 µs · ×1.541 99% CI 1.535–1.546 Regression
Bubble	1	7.146 µs IREGI 3.019 µs · ×2.367 99% CI 2.362–2.384 Regression	4.593 µs IREGI 3.020 µs · ×1.521 99% CI 1.516–1.532 Regression
Triangle	0	19.619 µs IREGI 4.975 µs · ×3.948 99% CI 3.932–4.027 Inconclusive: noise	11.310 µs IREGI 4.948 µs · ×2.281 99% CI 2.263–2.296 Regression
Triangle	1	26.724 µs IREGI 10.231 µs · ×2.610 99% CI 2.597–2.624 Regression	16.733 µs IREGI 10.279 µs · ×1.620 99% CI 1.608–1.649 Inconclusive: noise

12.1 Separately instrumented component timings



The stage sum belongs to a separate instrumented pass; "Other" is its total median minus named medians. These bars locate costs, not differences between revisions. Preparation is a single observation: bubble r0/r1 0.822/0.112 s; triangle r0/r1 0.067/0.061 s. It is excluded from the warm kernel ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs and all requested tensor slots. Structural allocations are reused; numerical results are recomputed. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`. All 391 source hashes, both binaries and the harness remain unchanged under the retained provenance checks. Absolute timings from earlier sessions cannot establish a revision speedup. Raw blocks, inputs, manifests and reproduction commands: `validation/benchmark/baseline-5c9e3dd/`.

13 Physical pipeline measurements: `80949db`

Six preserved-input binary64 comparisons pass complete canonical tensors against all three original modes and all three isolated graph Laurent slots against MadLoop. **Every measured timing row remains inconclusive**: its 99% interval exceeds the unchanged 2% relative-width threshold. Both top-pentagon rows fail during preparation; no output or timing substitutes for them.

Table 8: **Equivalent complete numerical pipelines, ms/point, TIRust / MadLoop.** Repeated captured point; amortized batch32 cost does not establish throughput over varying phase-space samples.

Graph	Ranks TIRust/ML	Batch 1	Batch 32
Top triangle	2 / 3	0.877 ms MadLoop 0.108 ms · ×8.114 99% CI 7.808–8.575 Inconclusive: noise	0.729 ms MadLoop 0.105 ms · ×6.918 99% CI 6.817–7.061 Inconclusive: noise
Top box	4 / 4	11.689 ms MadLoop 0.971 ms · ×12.036 99% CI 11.854–12.303 Inconclusive: noise	5.088 ms MadLoop 0.978 ms · ×5.203 99% CI 5.087–5.421 Inconclusive: noise
Massless up box	4 / 4	7.242 ms MadLoop 0.904 ms · ×8.008 99% CI 7.642–8.225 Inconclusive: noise	3.620 ms MadLoop 0.888 ms · ×4.075 99% CI 3.997–4.116 Inconclusive: noise
Top pentagon	4 / 5	Failed before timing Preserved-input conditioning guard	Failed before timing Preserved-input conditioning guard

13.1 Scope and uncertainty

The timers include numerical numerator coefficients, reduction, conversion, contraction, status checks and returned-buffer/cache cleanup. Graph/model and external-wavefunction preparation, input packing, final comparison adapters and process transport are excluded from both. Bare poles and R1 are retained; R2 and UV counterterm graphs are excluded. Routing and rank differences are declared algorithmic choices: **no tensor-stage timing ratio is valid**.

Both engines use Hybrid, 53 working/requested bits, input preservation, CPU 0, five warm-ups and 21 blocks targeting 30 ms. Separate engine runs use an independent bootstrap ratio of medians (10,000 resamples). These intervals describe within-run variation and do not bound drift between engines. Project computations and producer writes were frozen; external host activity was not controlled. Stage samples and raw full responses remain available.

Attempt002 preserves all source/build identities. Attempt001 is retained as unqualified because producer inputs changed during execution. The optimized top-box numerical pass is specific to this captured harness path; earlier CLI binary64 guard rejections remain part of the ledger. Top-pentagon failure is the actual preserved-input derived-distance conditioning diagnostic. No failure is removed from the family count.

Sources, contracts, stage timings and reproduction: `validation/benchmark/physical-joint-quiet/`. The preceding three-block ggH smoke and 80 Python boundary/provenance tests are separately retained in `validation/benchmark/physical-joint-smoke/`.

13.2 Precision follow-up on the failed pentagon point

Promoting the same 95 represented binary64 input components exactly to Float256/512, with requested accuracy53 unchanged, gives convergence over all 280 complex tensor slots within 2.17×10^{-55}. Original PV/hybrid agree within 4.62×10^{-14} and the preserved MadLoop graph within 9.29×10^{-14}. Original IBP still differs by 0.60013; binary64/128 guard failures remain. The first 256-bit driver captured valid outputs before a rational-input comparison-parser failure; the retained separate comparison passes. This is one-point manual precision qualification, not automatic rescue or a timing gate. See `validation/physical/gghhh-guard/`.

14 Current implementation checkpoint

Production snapshot `06706e2` and private experiments have separate source and binary identities. Correctness evidence does not replace timing gates.

14.1 Ordinary scalar-master storage

The default interpreter now reuses root temporaries through Symbolica's existing allocator. All five families preserve every output, precision and signed zero in 40 configurations each across f64, DoubleFloat and Float256. Shared helpers and Dual preparation are unchanged.

Root temporary slots	A0	B0	dB0	C0	D0
Before	38	503	1,951	220	25,010
After	6	13	24	8	25

The default-stack workspace snapshot compiled with unchanged source hashes before the generalized-PV promotion below. Its current-guard acceptance passes 30 execution, 29 stability, six public mixed-precision and three scalar integration tests.

The optimized `80949db` snapshot also passes eight complete bubble/ triangle kernel checks at batches 1/32, maximum normalized error 2.46×10^{-13}. Three short blocks, zero warm-up and concurrent builds make this a correctness smoke only; its recorded timing ratios are unqualified.

The paired DD E6 experiment preserves all three statuses and 12 complex Laurent outputs exactly, with the independent 10^{-23} check passing in both variants. Each uses order 128 and 3,720 D0 rows.

Diagnostic observation	Before	After
Whole validation duration	46.682 s	43.822 s
Per-child peak resident memory	2,052,724 KiB	558,752 KiB

These are single-run diagnostics, with other compilation active; no timing gate or speed ratio is inferred. The paired experiment uses an earlier, stricter Float guard; fixed-precision DD is unaffected. Current-guard tests are separate. Sources: `validation/stack-reuse/production-factory/`.

14.2 Generalized PV and unresolved precision

The retained original-IREGI dotted-integral comparisons cover positive propagator powers and dimension shifts; they are separate from COLLIER's ordinary undotted slice. Generalized PV for $N \leq 4$ passes nine focused tests and 36 original-IREGI comparisons across 312 complex slots; maximum normalized error is 7.04×10^{-12} at the unchanged 10^{-10} gate. The promoted PV and lazy-solve core also passes five supplemental behaviors: supplied products, multinomial weights, complex scalar/tensor comparisons and Hybrid isolation. The latter uses working128/256, requested53; its binary64 fixture failure is retained. These checks use frozen former-default master archives; combined default-stack integration is separate. Higher points retain IBP/E6. Sources: `validation/generalized-pv/` qualification, promotion and supplemental manifests.

The exact small-routing E6 oracle exposes a remaining production failure:

Working bits	Requested bits	Public result
128	128	Unstable: nonfinite box output
256	128	Unstable: nonconvergent quadrature

Both runs finish within their bounds and expose no successful values. No oracle comparison or requested-128-bit agreement is claimed. The frozen binary predates default stack reuse. See `validation/shifted-pentagon/production-series-qualification.json`.

15 Validation coverage

Table 9: **Evidence ledger.** “Source audited” records code inspection; “verified” identifies executed checks described in that row.

Capability	Required evidence	Status
Independent reference oracle	Untouched/repaired Fortran twins 24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass. New pentagon audits reveal reference defects.	Verified
Scalar-master component	OneLoopMaster A0/B0/C0/D0/DB0 Final27 scalar checks and13 consolidated D6 checks pass; all5 mixed physical D0 cases pass atFloat384. Unsupported derivative domains remain explicit.	Verified
IBP component	Powers, dimension shifts and singular identities 21 exact tests; the independent Gamma oracle covers 200 tensor requests and 40 generalized requests in all four slots.	Verified
PV component	Every tensor moment and native Laurent slot Canonical N1–4 ranks 0–7 pass. Generalized N≤4: nine focused + 36 original comparisons pass. Saved N5 rank-seven baseline: 1320 slots. Public N6/N7 tensor ranks pending.	Partial
Precision and Dual	binary64, DoubleFloat, Float, static/dynamic jets Selected f64/DD/Float128–512 and static/dynamic jets pass. Mixed-precision guards: 29 + 6 pass. Small-routing E6 requested at 128 bits: Unstable.	Partial
Execution / CLI / core oracle	Batches through reduction and scalar masters 30 execution and5 argument-guard tests pass. CLI19 smoke checks pass; frozen31 fixtures include30 binary64 plus1 explicit verified rescue.	Partial
Exceptional kinematics	Explicit errors and verified fallback N5 rank-six rescue passes; saved rank-seven baseline passes 1320 slots. Legacy: 22 primary + 8 rescues. E6 failures and broader domains remain visible.	Partial
Graph/numerator preparation	FeynKit DOT to Symbolica coefficients 4 model-boundary, 13 preparation, 4 physical numerator and 19 CLI smoke checks pass.	Verified
Physical MadLoop comparison	Graph-isolated top loops and massless box Integrated ggH, ggHH, ggHHH and massless up-box pass their retained MadLoop comparisons. ggHHH Float128/256 also agrees with original PV/hybrid; original IBP still disagrees and all-mode reports remain failed.	Partial

Snapshot boundary. Counts describe the executed component, public-execution and selected physical checks below. They do not certify all kinematics, higher-point ranks or every physical graph. Open discrepancies remain listed with the original tolerance. Four physical triangle and five massive/complex box cases pass six-component mixed derivative checks. Historical timeouts remain visible. The N5 rank-six public rescue verifies all 210 moments / 840 slots at 128/256 bits, including an independent coefficient. Binary64 TIRust and original IBP still exceed the fixed 10^{-10} criterion for that point. The frozen oracle passes 31 cases: 30 at binary64 and one generalized triangle through verified rescue. Broader domains remain pending.

15.1 Known original-reference defect

For shifts $(0,0,0,0)$ and $(0,1,0,0)$, squared masses $(4,9)$, $\mu = 3$ and numerator q_x^4, original PV and hybrid report stability but disagree with an independent Feynman-parameter integral. Their UV residues agree.

Method	Finite coefficient / i
Parameter integral	45.4316300015058951832997…
Original IBP	45.43163000150602
Original PV / hybrid	46.11123944517427

Evidence: `validation/pv-high-rank/evidence.json`. The original PV branch is retained unchanged; the relevant TIRust comparisons use original IBP and the independent parameter-integral result.

The dedicated original-IREGI/COLLIER table covers ordinary, non-dotted N2/N3 rank0/1 inputs, batches1/32. Its numerical and timing results do not qualify broader N4–N7, singular or IR-divergent families.

16 Measurement protocol

Pin release builds to one CPU core and record thread counts, hardware, OS, toolchain, flags, dependency/input hashes, seeds and warm-up. Identify cold or warm caches. Retain raw repeated-block observations, median, dispersion and an output checksum.

Preparation includes graph import, numerator construction and evaluator compilation. Time numerator, reduction, masters and total independently; summed instrumented stages are only diagnostic. Separate library costs from subprocess/file I/O and match physical routing, modes and stage boundaries.

Use identical samples at batch widths 1, 4, 8, 16, 32 and 64, matching numerical precision and including rescue costs. Retain failed lanes. Every speed gate requires correctness: stable binary64 uses the fixed 10^{-10} mixed tolerance on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput per topology/rank/mass family, with IBP, PV and hybrid separate. The target envelope includes seven propagators, rank seven, generalized powers and dimension shifts.

The kernel harness now records every requested configuration before starting subprocesses, preserving interrupted and unrun rows. The latest Python suite passes 80 ledger, provenance, protocol and comparison checks. The optimized joint ggH smoke passes at batches 1/3; its short zero-warm-up timings are diagnostic. The subsequent quiet physical results are reported separately.

Private slim RealBall passes 88 core checks, eight kernel and 15 typed cases, two ggH packets, and 10 separate planar-interaction checks with retained dependencies. Its first four timing rows give two noisy real-lane results, one complex-lane parity and one gain. The longer cohort is reported separately; the candidate remains unadopted.

With the accepted license, 360 batched calls across two workers match sequential A0–D0 values, status and metadata for f64/Float128 and second-order Dual branches. A separate four-worker cold-specialization test passes shared ownership and cleanup checks. These qualify retained SymbolicaBatch snapshots; native kernels remain separate. The earlier rank-one box rejection is retained in `validation/concurrent-workspaces/`.

17 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `bash scripts/build-benchmark-report.sh`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst's `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

17.1 Validation reproduction and remaining checks

Run recorded commands with the pinned environment from `scripts/env.sh`. Exact commands and source hashes accompany each JSON result. The frozen 31-case oracle includes one explicit 128/256-bit rescue; all 30 legacy fixtures pass through 22 primary results and eight verified rescues. Three active rescue, four bubble-Taylor, three scalar API and nineteen CLI smoke checks pass. The new 68-test default-stack acceptance is a separate source scope; earlier failures remain retained.

The final frozen master source passes 27 checks, including extreme-scale B0, six-component scale-only jets in all families, mixed input precision and the full physical D0 derivative batch; see `validation/scalar-masters/qualification.json`. The separate exact factory-parity/default-stack qualification is described above. Generated native kernels remain opt-in; no new timing gate passes.

An independent Symbolica Gamma expansion checks 200 tensor and 40 generalized requests with analytic bubbles disabled/enabled: 480 evaluations and 1920 slots. Nine extracted FeynKit partial-fraction and two planar formula tests pass; the separately qualified explicit planar API is recorded below.

N6 inline 128/256-bit arrays converge over 1320 slots within 6.46×10^{-29}; N6 rank7 and N7 arrays remain unqualified. The later restricted public N6 rank2/3 qualification is recorded separately below. Adopted alias-use collection preserves 29 paired cases, 86 successful lanes and 4520 components. Frozen collector-plus-sum and private amplitude-variable probes reach the 600 s bound (601.232/601.329 s with teardown), after primal amplitudes but before master activations finish; neither produces arrays. The latest private variant passes 48 variable-set, 34 amplitude/error and 73 semantic checks. Earlier timeouts remain in `validation/public-highpoint/` no timing gate follows.

18 Physical fixture qualification

Single graph	Pointwise numerator	Integrated TIRust result
Top triangle: $gg \to H$	Verified	Verified
Top box: $gg \to HH$	Verified	Verified
Top pentagon: $gg \to HHH$	Verified	PV/hybrid + MadLoop
Massless up-quark box	Verified	Verified

18.1 Physical reference contract

Fixtures preserve parameters, actual HELAS polarizations, graph factors and source hashes. The triangle uses two orientations with grouping disabled. MadLoop shifts are $(0, p_1, p_1 + p_2)$ with polynomial variable $x = -Q$; its adapter includes the odd-rank sign. For the FeynKit routing, $N_{\mathrm{FK}(q)} = -\frac{1}{2} P_{\mathrm{ML},1}(p_1 - q)$ with matched couplings/polarizations. Do not apply color or direction factors twice.

Five captures contain 40 graph records, including a second triangle helicity; these are not 40 integrated validations. Thirteen preparation and four selected physical numerator comparisons pass. Actual evaluated momenta are exported because MadLoop may correct the requested point.

Fresh preserved-input Fortran replay matches all 301 numerator coefficients, 903 tensor Laurent entries and 12 contracted slots across four graphs exactly; eight alignment boundary tests pass. Legacy-default ggH is unchanged. This is reference-only qualification; later TIRust checks belong to the separate joint run. Equivalent full-graph timing retains TIRust/MadLoop ranks 2/3 for ggH, 4/4 for both boxes and 4/5 for ggHHH, and their declared routing. It does not qualify tensor-stage ratios. Sources: `validation/benchmark/physical-alignment/`.

The recorded binary64 $gg \to H$ CLI passes all 15 moments / 60 slots against original IREGI within 1.86×10^{-15} normalized error and MadLoop within 9.38×10^{-15}, including poles. Values and binary identity remain in `validation/physical/ggh-cli-comparison.json` this is not a fresh replay. Nineteen CLI smoke checks pass. Binary64 comparison arithmetic certifies no extra digits from higher working precision; short unpinned timings are excluded.

The historical top-box binary64 CLI rejects: PV fails its cancellation guard and IBP fallback is nonfinite. Float128/256 reruns pass all 280 slots against all three reference modes within 1.15×10^{-13}, and MadLoop within 4.77×10^{-15}. Each freshly prepares/evaluates the numerator. Their normalized tensor difference is 6.09×10^{-37} under binary64 comparison arithmetic. See `validation/physical/gghh-cli-precision-convergence.json`. The Symbolica-backed model adapter passes four boundary, thirteen preparation and four numerator tests while preserving other numerical tokens.

The 80949db massless up-quark box passes all 70 moments / 280 slots against each original mode: maximum error 2.85×10^{-14} against IBP and 7.11×10^{-15} against PV/hybrid. MadLoop finite/single/double coefficients agree within 6.74×10^{-15} with fixed-color factor $-\frac{1}{8}$. Its one-run wall time is unqualified; see `validation/physical/gggg-up-cli-comparison.json`.

Complete top-pentagon CLI reruns at Float 128/256 now agree with original PV/hybrid for all 280 slots within 4.63×10^{-14}, and with MadLoop within 9.30×10^{-14}. Between-run tensor difference is 7.79×10^{-17} under binary64 comparison arithmetic. Original IBP still differs by 0.60014, so the all-mode report remains failed. See `validation/physical/gghhh-cli-precision-convergence.json` and linked reports. Earlier PV/hybrid replay of 378 captured MadLoop slots was exact.

18.2 Physical derivative qualification

Four triangle cases at 384 bits cover real subthreshold/absorptive and complex-mass kinematics, each with six first, second and mixed Dual components. Finite differences converge with maximum final scaled error 5.47×10^{-28}. One equal-complex D0 case also passes, with error 1.11×10^{-34}. The complete five-case D0 batch now passes at 384 bits, maximum final scaled error 2.53×10^{-30} under the unchanged 2^{-70} criterion. Its 224.83 s validation duration is not a benchmark; earlier 180 s timeouts remain retained. See `validation/scalar-masters/physical_box_full.json`. Exact commands and scope are retained in `crates/iregi-masters/tests/validation_evidence.json`.

19 Open qualification ledger

Unresolved item	Observed evidence and remaining check
Binary64 baseline	Historical TIRust/IBP errors 2.23×10^{-10} / 3.40×10^{-10} fail. Rank-six rescue passes at the retained point.
Pentagon coverage	Rank-six rescue passes 840 slots; saved rank-seven baseline passes 1320. Later precision/routing edits and broader domains require qualification.
Physical pentagon	Float 128/256 agrees with PV/hybrid and MadLoop; original IBP disagrees. Binary64 and all-mode acceptance remain incomplete.
Shifted scalar	Thirteen standalone checks and public f64 fixtures pass. Production small-routing E6 remains Unstable at requested 128. New private mixed-jet checks cover common masses/zero invariants only; broader domains and production integration remain unqualified.
Derivative domains	Five-case mixed D0 checks pass 2^{-70}. Thresholds and unsupported transverse massless/Gram/Cayley domains retain errors.
Physical top box	Historical binary64 guard rejection remains visible; Float 128/256 and preserved 80949db binary64 checks pass at retained points.
Supplied-product N6	Certified scalar working 128/256, requested 53 passes with zero poles and invalid-lane isolation; binary64 rejects. Independent difference 1.53×10^{-23} is below estimated (nonrigorous) 6.11×10^{-22}. N7/tensors/mixed Dual remain pending.
Legacy and precision	All 30 retained cases pass: 22 primary and 8 rescues. Three false-success cases are repaired; conservative active-master error estimates remain noncertified. Broader domains remain unqualified.
Performance	Historical 4c32 interpreter: 7 regressions/1 gain; native: 5 regressions/1 parity/2 gains. 5c9: 6 regressions/2 noisy; 80949: 5/3. Latest 6b24: 7/1. Physical: six noisy/two failures. Overall acceptance is unmet.

19.1 Optimization and experimental evidence

Candidate	Executed evidence and decision
Adopted allocation / guards	Core allocation: 67 current opt2 checks; quiet paired run preserves 92 outputs. Guard context: 76 checks and 120 snapshots, 14 cleanup checks; six isolated gains/six noisy rows. Combined 6b24 optimized IREGI gate remains seven regressions/one noisy row.
Roundoff cache	29/37 stability checks and 120 snapshots pass. Candidate rejected: two regressions/ten noisy rows. Separate accuracy-guard diagnostic: two regressions/six noisy rows.
CLI coefficients	72 ordinary/12 mixed batches pass. Eight opt2 timing rows: five gains/one parity/two noisy; ggHH32 ratio 1.0367, CI 1.0237–1.0501 retains possible regression; unadopted. CSE proves 892 coefficient/223 homogeneity identities. 1281 raw failures coexist with zero dimension-normalized diagnostic failures; no physical acceptance or new timing claim.
Native candidates	Dense f64 B0/C0: two gains/six singleton parities/four batch32 regressions. D0 allocator: three singleton gains/two batch32 regressions/one noisy. Shared B0/C0 kernels: one gain/two parities/seven regressions/two noisy. All rejected; D0 singleton source is unchanged. Separate actual A0–C0 replay passes 1782 outputs from 297 rows at 10^{-10}; native/interpreter values are not bitwise identical. New replay timings are reported separately.
Certificate candidates	Slim RealBall module: 23 checks, including 8 baseline controls. Separate exact-distance receipts: 49 checks and 120 exact snapshots against the same private RealBall baseline; argument rows unchanged. Fourteen timing cases give eight gains/two parities/four noisy, no qualified regression. Private final integration passes 12 receipt/6 accounting checks, counted once with the automatic-planar followup. The final timing cohort is reported separately.
N5 planar API	Promoted explicit unit-power API: rank7 ordinary128/256 passes 1320 slots/convergence; mixed128 passes 5280 slots per accepted lane versus256 boxes, requested53. Transverse jets reject; 9 focused/49 ordinary checks and 20 Fortran-leaf coefficient cases pass. Earlier timeout remains. Private automatic rank1/2 dispatch passes 69 checks; the final 18-test followup is separate. No automatic rank7, production fallback or timing claim.
Finite D0 certification	Explicit API promoted: three actual public tests and current core Cargo checking pass. Exact represented inputs receive component-budget enclosures in the checked near-vacuum domain; upstream routing and quadrature errors are excluded. Prior typed targets are f64/40, DD/80, Float128/100 and Float256/128. Ordinary evaluation and default E6 remain unchanged; no automatic certified-master dispatch.

20 Latest isolated optimization cohorts

These private candidates were measured in one coordinated quiet window on CPU 254. Each has its own frozen sources, binaries and registered protocol; results do not qualify a combined implementation or replace the TIRust/IREGI performance gate. Candidate effects are never multiplied. Failed controls, inconclusive rows and earlier cohorts remain in the evidence.

Private cohort	Per-family outcomes	Controls, scope and decision
Final proof receipts	14 families: 11 gains, 1 parity, 2 regressions; no noisy intervals.	Ordinary bubble singleton regresses 2.90% (CI 2.43–3.27%); ordinary triangle batch32 regresses 0.69% (CI 0.20–1.26%). No blanket adoption. Same retained masters; N5 path inactive. Earlier cohort remains separate.
Slim RealBall, longer run	4 families: 1 gain, 1 noisy, 2 regressions.	Real singleton improves; real batch32 remains noisy (CI width 2.18%). Complex-mass controls regress 0.74%/0.62% at batches1/32. Real shifts remain; no adoption or pooling with the first run.
Native master replay	8 families: 8 gains; all 168 pairs faster.	TIRust(candidate)/TIRust(baseline) ratios 0.143–0.809 on actual A0–C0 master bundles: native candidate, Symbolica baseline. Raw kernels exclude adapter validation/allocation, recurrence and contraction. No production switch or IREGI parity claim.
CLI coefficient CSE0	8 families: 8 gains.	Complete warm coefficient call only. Raw coefficient criterion still fails in baseline and candidates; separate dimension-normalized diagnostic passes. No adoption or physical acceptance claim.
CLI coefficient pair1	8 families: 6 gains, 2 noisy.	ggHHH singleton and up-quark box batch32 remain inconclusive. Preparation observations are retained separately; repeated fixed-point f64 evaluation is not a full-graph timing gate.

Ratios in these cohorts are **TIRust(candidate) / TIRust(baseline)**, using each cohort's own implementation and baseline. Every measured family is retained in the linked JSON, with raw paired observations, 99% confidence intervals and unchanged qualification thresholds. Project compute was held, but unrelated host activity remained uncontrolled. Earlier measurements use different frozen sources or protocols and remain separate. No new production adoption is implied by this page.

20.1 Exact and source-only research

Additional Symbolica CSE leaves A0/B0/C0 exports unchanged: 43/506/223 instructions and 6/13/8 root slots. Eighteen root identities and 470 control checks pass under the structured-SSA assumption. With no instruction saving, the candidate stops before numerical or performance qualification.

The E6 design notes remain proposals. The initial private direct N2 prototype at Float256/request128 certifies 16 components across four successful lanes with one-axis common-mass jets; five expected rejections hide values. Independent audit: 513 checks and 140 hashes pass. Its later typed followup appears separately below. Neither snapshot qualifies public/default E6, general geometry or performance.

A separate private rank-lowering N6 rank-one candidate passes all 20 complex Laurent slots against original IBP at Float128/request53, maximum normalized error 3.17×10^{-13}. The old public path rejects nonfinite coefficients, so this is not paired old/new numerical agreement. Exact constraint tests accompany the frozen development build. Its rank-two probe rejects nonfinite coefficients in 57.823 s, within the 180 s bound, without arrays. Rank-seven, N7, convergence and performance qualification remain pending.

21 Lazy proof capture: complete paired comparison

TIRust(lazy) / TIRust(eager) measures the entire reduction call, including result construction and disposal. Variants share the eager-final baseline, current master adapter and unchanged optimized numerical dependencies; only receipt-capture allocation changes. Symbolica masters remain selected; N5 retry is inactive.

Table 10: **All fourteen workloads are retained.** Ratios below one favor lazy capture. Each cell shows the paired latency ratio, its 99% bootstrap interval and the unchanged per-family verdict.

Workload	Batch 1	Batch 32
Bubble · mixed Dual	0.9945 99% CI 0.9932–0.9969 Gain	0.9996 99% CI 0.9958–1.0021 Compatible with parity
Bubble · ordinary	0.9913 99% CI 0.9871–0.9976 Gain	0.9957 99% CI 0.9935–0.9968 Gain
Bubble · complex / exact zeros	0.9972 99% CI 0.9953–0.9979 Gain	1.0008 99% CI 0.9987–1.0031 Compatible with parity
Bubble · real / exact zeros	0.9984 99% CI 0.9940–1.0045 Compatible with parity	1.0089 99% CI 1.0038–1.0159 Regression
ggH · mixed complex	0.9979 99% CI 0.9957–1.0019 Compatible with parity	0.9999 99% CI 0.9975–1.0017 Compatible with parity
ggH · real / exact zeros	0.9951 99% CI 0.9890–0.9973 Gain	1.0022 99% CI 1.0002–1.0061 Regression
Triangle · ordinary	1.0077 99% CI 1.0053–1.0108 Regression	1.0018 99% CI 0.9981–1.0034 Compatible with parity

Five gains, six parity results and three qualified regressions; no noisy intervals. No adoption. Regressions occur for the real exact-zero bubble at batch 32, real exact-zero ggH at batch 32, and ordinary triangle singleton. Every 99% interval meets the fixed 2% width gate. The all-family no-regression criterion fails; all families remain retained.

Fourteen optimized preflight pairs preserve 10,560 unique complex output slots and exact status/metadata/context/counters. The earlier 26 debug tests remain separate. CPU 254; five warm calls; 41 paired blocks with 21/20 alternating order; retained calibration ≥200 ms on the faster side; 20,000 paired bootstrap resamples; all 1,148 observations retained. Lane/output checks precede and follow timing; iterations unwrap the top-level result. The host is shared.

Scope: N2/N3 f64 and flattened mixed Dual only; no new TIRust/IREGI or TIRust/COLLIER gate. Earlier eager/core and slim-RealBall cohorts remain separate; their ratios cannot be multiplied. Source: `validation/lazy-receipt-performance/`.

22 Additional bounded correctness checks

These completed checks extend the evidence ledger without changing production or any measured timing row. Each private candidate retains its own source and dependency snapshot; the raw-master replay gains do not establish whole-reduction performance for a different candidate.

Bounded study	Observed result and remaining limit
Selected native A0–C0	Eight ordinary f64 N2/N3 rank0/1 families at batches 1/32 pass all 1584 complete tensor coefficients per backend against original IREGI and the Symbolica baseline. Maximum normalized errors are 3.38×10^{-13} against IREGI and 2.07×10^{-13} between backends. The 297 actual master rows retain all 1782 six-output coefficients. Private opt0 build; whole-call performance is unmeasured.
Lazy proof capture	26 tests pass: 12 receipt, 8 lifecycle and 6 accounting checks. Fourteen paired workloads match exact values, status, metadata and counters through seven calls per implementation. Capture is created only when evidence is needed and discarded after each call. This private debug qualification is separate from the later optimized timing table; no adoption.
COLLIER N4 extension	Eight fixed massive cases cover ranks0–3 at batches1/32. Both scalar rows are accepted, maximum normalized error 2.76×10^{-14}. All six higher-rank COLI rows retain `AccFlag=-1`, despite raw agreement within 2.39×10^{-12}. DD mode returns `ErrFlag=-10` for all eight; those arrays are invalid reference evidence. No inputs or thresholds changed; no timings were run.
Fused core CSE	The N2 rank1 native/OneLoop exports remain byte-identical: 1013/850 instructions and 237/147 slots. All 422 exact roots and 730 control checks pass under the structured-SSA assumption. With no structural saving, the candidate stops before numerical or timing work; other topologies remain untested.

22.1 Scope of the independent checks

The selected-native study also passes 34 adapter cases, ten unchanged master paths covering 33 cases and three unchanged whole-core paths. Repeated whole responses are retained; direct adapter repeats are executable assertions with the common output saved. Initial build and fixture failures remain in the package. Native D0, high-point tensors and a production backend switch are outside this qualification.

The lazy-capture allocation occurs inside the existing argument-stage timer. The earlier eager allocation was outside that stage; the later paired comparison therefore uses total latency. The earlier final receipt cohort and its two qualified regressions remain unchanged.

N4 checks use 238 Symbolica-prepared polynomial arrays for metric traces, pinch translation, denominator-difference signs, UV finiteness and odd-time reflection. Their largest identity discrepancy is 2.89×10^{-12}; these constraints do not independently determine every spatial tensor component or override a rejected library status. The initial eight measured IREGI/COLLIER rows remain unchanged.

A retained **initial unmasked** thin-N6 rank2 probe passes seven exact/sidecar checks, then both certified-planar retries fail among six packed N5 children; that attempt produces no parent arrays. The later private mask followup has separate source, numerical and lifecycle qualification below. Those private capsules establish no production or performance gate; the later explicit public six-point API has its own qualification below.

Historical compile/type/lint checks appear separately from later wheel004/gallery execution. Binding and notebook sources follow the pinned Community guide in `PLAN.md`.

Source bindings, raw outputs, failures and reproduction commands are retained in `docs/data/results.json`. No elapsed correctness-run duration on this page is a benchmark. These historical records bind their named source revisions; the optimized TIRust performance gate remains the measured `6b24c03` snapshot, without the later six-point API changes.

23 Additional execution qualification

23.1 Private N6 rank-two mask followup

Fourteen exact checks qualify the thin child assembly, full-shape zero-mask certificates, cache/context binding and route envelopes. Float128 and Float256, both requested 53, return all 60 native complex Laurent slots for the retained base point. Maximum normalized disagreement with original IREGI is 1.165×10^{-12} against the unchanged 10^{-10} gate; maximum 128/256 difference is 1.474×10^{-34} against 2^{-80}.

Check	Retained count	Scope
Precision comparison	2 × 60 slots	One retained base point at Float128/256; requested 53.
Lifecycle	11 calls / 17 accepted lanes	1020 complex slots; seven explicit rejection controls and two empty resets.
Singleton agreement	Zero differences	Values, precision and signed zeros agree with separate singleton calls.
Independent audit	327 bindings / 1140 slots	Saved-data arithmetic and source-archive audit; no kernel rerun.

The lifecycle corpus includes base/scaled points, mixed valid/invalid batches, reordering, chunks and resets. It shows isolation and absence of partial successful arrays; a rejected status alone does not prove the mathematical failure cause. The ordinary primal sensitivity estimate remains nonrigorous for compounded Taylor errors, and imported quadrature errors are unsupported.

Strict scope: private FromShifts/Preserve, unit powers, no dimension shift, N6 rank 2, native raw components. Numerical Dual, complex masses, N7, rank 7, generalized inputs, production adoption and performance remain unqualified. The earlier unmasked rejection and the 30 s exact-test timeout with zero completed tests remain preserved. The later unchanged 14-test gate passes within its separately declared 180 s bound. Sources: `validation/highpoint-thin-n6-mask/` and its independent checkpoint audit.

23.2 Private generic rank-three extension

A separate generic assembly passes 25 exact structural/integrand/context tests. At the retained N6 rank 3 base point, Float128/256 with requested 53 each return all 140 native complex Laurent slots, including 105 exact zero poles. Maximum original-IREGI discrepancy is 2.426×10^{-12} against 10^{-10}; precision difference is 1.474×10^{-34} against 2^{-80}. The exact tests cover structural identities through rank 7; numerical rank 7 is not qualified. This separate package adds no rank 3 lifecycle, numerical Dual, generalized/SuppliedG, N7, public adoption or performance result. The initial test compile failure and historical rank 7 timeouts remain. Source: `validation/highpoint-thin-n6-generic/`.

23.3 Initialization-order diagnostic

Two fresh processes retain 55 complete responses: seven control requests and 48 case requests. All numerical admissions pass; 37 within-process exact comparisons have zero differences. CASE-first reproduces all 32 saved case responses and all three controls exactly. CONTROL-first reproduces a one-ULP C0 difference and changes 24 finite imaginary tensor values per backend in the batch 32 rank-one triangle, by at most nine ULP (2.50×10^{-16}). The whole-case master captures and input/status/diagnostic records remain exact.

Registration/operand-order source facts suggest a mechanism, but no paired instruction exports establish the cause. This is an untimed, schedule-associated diagnostic, not universal determinism or an accuracy-threshold change. The failed v2 preflight remains unmeasured. A later four-engine cohort uses its separately reviewed CASE-first protocol; no diagnostic duration is a benchmark. Source: `validation/selected-native-initialization-order/`.

23.4 Python interface: historical compile and source typing

The PyO3 standalone/native crate passes Cargo checking with one shared Symbolica/Numerica/Graphica/PyO3/FeynKit owner identity. Its ordinary source consumer passes basedpyright with zero errors and warnings. This checks compilation and source-facing type signatures; installed-wheel typing and execution require the separate runtime evidence below. The initial build timeout and retained mypy/upstream-stub failures are preserved rather than counted as successful checks.

The initial notebook source capsule retains its original strict lint and nine display-helper tests. Later explorer, browser and native notebook checks are separately source-bound; historical results do not qualify a later source revision. Sources: `validation/python-interface/checkpoint-compile/` and `validation/python-notebooks/qualification.json`.

24 Historical Python and notebook runtime

24.1 Inspected native wheel baseline

The exact baseline wheel builds, installs and passes shared-owner identity, packaged font/stub checks and installed-consumer typing. Of 29 focused tests, 28 pass; the real upstream graph renderer fails with `painted coordinates must be numbers`. Numerical, precision, Dual, dotted, batch, contraction and shared tensor/graph interoperability checks pass independently of that rendering failure. Tests use the actual shared upstream types from the installed native host.

Release-wheel performance, general mathematical-domain acceptance and full Community-host integration remain unqualified. Historical `6b24c03` measurements describe an earlier kernel. Source: `validation/python-interface/checkpoint-wheel-003/qualification.json`.

24.2 Actual native notebook execution: partial result

The inspected wheel is installed into the marimo 0.24.0 notebook environment. Four native test methods complete: two pass, two have three rendering-error occurrences. The successful controls check the dotted bubble against its mass Dual derivative, precision distinctions beyond binary64, failed-lane isolation and an edited-topology guard. Finite-difference values are displayed; this test adds no independent finite-difference acceptance gate.

Notebook	Actual checks	HTML export	Qualified scope
01: Graph → integral	Real renderer errors	21 cells / 1 error	Bubble, triangle Float and bubble DoubleFloat stop before their numerical assertions.
02: Dots and derivatives	Numerical controls pass	19 successful cells	Massive bubble identity, exact-input digits and invalid-neighbor masking.
03: Benchmarks	3 tests / 6 cohort choices	11 successful cells	Frozen evidence only; no numerical extension imported.

Both native exports and the original error remain archived. Rendering-dependent numerical assertions were unrun in this baseline. The later wheel004 followup below qualifies the named examples and their static graph/math browser displays. Source: `validation/python-notebooks/native-wheel003/`.

24.3 Benchmark explorer and readable narrow charts

The separate explorer source revision passes strict lint and 15 evidence/display-helper tests, including all forty saved ratios and eighty CI endpoints. Actual marimo execution retains every cohort and its original short/noisy gates. Static Chromium desktop/narrow-dark checks preserve eight ratio cells and render without request, HTTP, console or page failures.

The chart now keeps a 930 px SVG inside controlled horizontal overflow, preserving readable labels within a 348 px narrow viewport; the document stays 390 px wide. Desktop cells and narrow/dark grid/chart crops were visually inspected. Earlier loader failure and small-chart screenshots remain historical. The export requires the official marimo CDN frontend and has no reactive backend; browser snapshots do not qualify numerical controls. Sources: `validation/python-notebooks/chart-readable/` and its separate source/runtime/browser predecessors.

25 Repaired Python interface and executed gallery

25.1 Exact JSON and actual upstream graph rendering

GDB inspection of the unchanged baseline wheel finds that exact JSON's `arbitrary_precision` feature turns coordinates serialized through `serde_json::Value` into CBOR private-number marker maps. The actual numeric-coordinate decoder rejects those maps. The pinned upstream owner now uses typed line/cubic command tuples and typed collision-line records at both serialization sites. The exact-JSON feature and actual renderer remain in use.

Two earlier owner-test setup/fixture failures remain archived. The final two consumer regression tests pass, exercising both stroke decoding and collision search under exact JSON. Rebuilt local wheel004 then passes all 29 actual focused tests: 17 numerical and 12 interoperability, including real graph rendering, with no failures or skips. Shared-owner identity, packaged assets/stubs and installed consumer typing pass. This is a local development wheel; build durations are validation diagnostics, not timing comparisons. Sources: `validation/python-interface/checkpoint-wheel-004/qualification.json` and `validation/python-interface/rendering-diagnosis-003/`.

25.2 Unchanged notebooks, fresh installed wheel

The approved protocol freezes eleven notebook/test/graph/environment inputs plus the exact wheel and owner receipts. A separate fresh environment preserves both the failed baseline003 environment and the inspected owner004 environment. All four native test methods pass. They exercise bubble f64, bubble DoubleFloat and triangle Float graph preparation, full tensor output shape, batching and contraction, plus the dotted bubble/Dual mass identity, precision distinctions, invalid-neighbor isolation and the edited-topology guard. Finite-difference values are computed for the display; this test adds no separate finite-difference accuracy threshold.

Notebook	Native / evidence checks	Successful HTML cells	Real static browser display
01: Graph → integral	Graph/numerical controls pass	21 / 21	Actual upstream bubble SVG, formulas and four contracted Laurent cards.
02: Dots and derivatives	Identity/precision/ isolation pass	19 / 19	Formulas and four dotted-integral Laurent cards.
03: Benchmarks	3 tests / 6 cohorts	11 / 11	Eight ratios, retained short gate and readable horizontal chart.

All three strict lint checks and 15 display/evidence tests pass. One Chromium pass inspects each export in desktop light and narrow dark views: six views, 1,918 normal frontend requests, no failed requests, HTTP, console or page errors. The mobile document stays 390 px wide; the 930 px chart pans inside its 348 px viewport. Actual graph, numerical cards and chart crops were visually reviewed. The static export uses marimo's official CDN frontend and has no reactive numerical backend. No numerical mock or replacement renderer is used. Release/platform/Community-host and full mathematical or performance qualification remain open. Source: `validation/python-notebooks/native-wheel004/qualification.json`.

26 Explicit public six-point reduction

The source now exposes `SixPointReductionPlan` and `SixPointWorkspace<T>` as a separate, restricted API. It accepts six unit-power propagators, ranks 0–3, dimension shift zero, preserved Cartesian shifts, PV mode and native IREGI raw components. Unsupported settings fail during preparation. Ordinary reduction, hybrid dispatch and CLI defaults retain their existing behavior.

Isolated qualification	Retained count	What was checked
Selected exact tests	34 passed	Assembly, context, certificates and API controls in the frozen isolated build.
Existing ordinary API	12 paired cases / 2200 slots	42 successful lanes and 18 matching failed controls; values, precision, signed zero, metadata and counters preserved.
N6 rank 3	140 slots × 2 precisions	Float128/256, requested accuracy 53; all 105 pole slots are exactly zero at each precision.
N6 rank 2 lifecycle	11 calls / 17 accepted lanes	1020 complex moments and 68 contractions; seven rejection controls and two empty resets.

The rank-three base point agrees with original IREGI within 2.426×10^{-12} under the unchanged dimensionally normalized 10^{-10} gate. The 128/256-bit difference is at most 1.474×10^{-34} against 2^{-80}. The rank-two lifecycle retains mixed valid/invalid batches, scaling, reordering, chunks and post-error reuse. Its contraction check preserves the requested native convention and tensor metadata; it does not require bitwise equality after reassociation.

The implementation packs all six five-point children across active parent lanes, preserves child sensitivity diagnostics and checks complete Taylor cofactor constraints before dropping zero contributions. Failed lanes expose no partial successful coefficient array. These are implementation properties covered by the named tests, not a rigorous numerical error enclosure.

Separate current-Cargo replay. The actual locked non-Python Cargo library test and example builds pass, followed by all 34 selected tests. At the retained rank-three point, both complete 140-slot arrays pass the unchanged original-IREGI and precision-convergence gates. Independent saved-data arithmetic verifies all 280 original comparisons and 140 convergence checks. The isolated qualification remains unchanged; its 12-case ordinary comparison and rank-two lifecycle are not reported as additional current-Cargo runs.

Provenance binds the actual linked executables and 674 runtime inputs, including recorded Cargo artifacts and Rust source guards. This is not a complete native build-input closure: three native archives have supplemental post-run hashes only, without a retroactive pre/post guard claim.

The accepted high-point numerical scope is the retained ordinary Float rank2/3 corpus at requested accuracy 53. Public N6 Dual, binary64/DoubleFloat, arbitrary complex masses, generalized powers/dimension shifts, rank 7, public N7 and performance remain unqualified. Existing ordinary Dual controls above preserve the old API; they do not qualify high-point Dual evaluation. Earlier failed high-point attempts and original-IREGI rank7 discrepancies remain visible. Sources: `validation/highpoint-six-point-api/qualification.json` and `validation/highpoint-six-point-current-cargo/qualification.json` public usage: `docs/six-point.md`.

27 Current-wheel C0/D0 mass-Taylor identities

The unchanged installed local wheel004 now passes first, second and mixed squared-mass Taylor identities for rank-zero C0 and D0. Each family uses one fixed massive spacelike point, working precision 128 and 256 bits, and requested accuracy 53. The ordered six-component Taylor layout is `[[0,0],[1,0],[0,1],[2,0],[1,1],[0,2]]`.

Family	Successful evaluations	Laurent identity checks	Precision checks	Maximum normalized identity error
C0	14 / 14	48 / 48	48 / 48	2.509×10^{-36}
D0	14 / 14	48 / 48	48 / 48	3.712×10^{-36}

All 28 evaluations succeed. The 96 identity checks compare the Taylor coefficients with independently prepared raised-power IBP reductions at the appropriate factorial normalization. Another 96 checks compare the two working precisions. Every check includes native UV single, IR single, IR double and finite coefficients; all three poles are exactly zero at these points. The finite coefficients remain general complex values. The saved-data audit checks 192 evaluated complex Laurent slots through 9238 assertions using exact integer ratios and the original dimensionally normalized 10^{-10} criterion.

Both families use the existing native convention and preserved inputs. No fallback or analytic rescue occurs. There is one numerical attempt per family and no retry. The original C0 saved-data audit incorrectly required a zero real finite part; its failed receipt is retained. The corrected audit checks the full complex value, with unchanged numerical output, input, pole tests and tolerance. No new integral evaluation was used to repair it.

Evidence identity. This is wheel004 evidence, bound to native module `cd4df512a134…` and its saved source archive. It is independent of the later public six-point source/Cargo check. The raised-power comparison and independent saved-data arithmetic are not a new external-library or finite-difference validation.

Working at 128/256 bits does not claim that many achieved accuracy bits: the requested target is 53. These two cases do not add general static-Dual or DoubleFloat coverage, massless or threshold derivatives, collinear/Gram/ Cayley boundaries, arbitrary complex-mass domains, further sample batching, physical/MadLoop comparison or performance qualification. Historical broader C0/D0 finite-difference evidence remains a separately bound source scope. Source: `validation/c0d0-dual-identity/qualification.json`.

28 Private capability and retained high-point failures

28.1 Selective native-f64 masters: correctness and metadata

The private opt-in A0/B0/C0 backend passes 17 focused tests and a retained 38-request capture. Per backend, the whole N2/N3 rank0/1 batch1/32 fixtures contain 1,584 complex tensor outputs and 1,782 complex outputs across all six master coefficients. The largest candidate/original dimensionally normalized error is 3.38×10^{-13} against 10^{-10}. Separate selected, unchanged precision/Dual, cache and failure controls remain visible in the package.

The metadata compatibility matrix now has all 16 entries passing across default, selective native, broad native-batch and combined features. The independent continuation completes only the two unfinished metadata checks; earlier timeouts, cancellation and the false final guard remain historical. Metadata checking does not execute the broad/combined numerical test binaries. No production/default change, performance qualification or adoption follows. Sources: `validation/native-f64-opt-in/capability-v1/qualification.json` and `validation/native-f64-opt-in/compatibility-completion-v1/qualification.json`.

28.2 Uniform-branch scheduler: local fixture qualification

The private Symbolica scheduler candidate passes 15 actual tests: five existing batch, six focused and four nested/shared-function tests. Actual old-batch, candidate and scalar paths are compared, including precision, signed zero and active instruction/lane traces for f64, DoubleFloat, mixed Float precision and static/dynamic mixed-order Dual shapes. Two earlier fixture compile failures remain archived. The patch is not applied to live Symbolica; dependent masters/core, full reduction and performance are not qualified by these local tests. Source: `validation/uniform-branch-scheduler/qualification.json`.

28.3 First private N7 scalar capture: no Rust numerical array

The retained dyadic N7 rank0 fixture passes its exact-domain audit, including seven six-point determinants and 21 child coordinate ranks. Its generic structural plan times out at `preparing_plan`, before workspace or scalar evaluation. There are zero Rust numerical slots. Cleanup completes, but total capture time including guards is 300.928 s, exceeding the frozen 300 s bound; the comparison is not spawned.

Original IBP and hybrid report stable matching finite values; PV is unstable and its zeros remain rejected. An independent scalar oracle finishes all six fixed rules (orders 8/12/16, two mappings), but its empirical normalized uncertainty estimate 7.696×10^{-10} exceeds 10^{-10}. That oracle exits with insufficient convergence. No numerical agreement, precision convergence, general N7, higher rank, Dual or performance acceptance follows. The earlier 17 selected exact tests remain a separate structural result. Source: `validation/highpoint-thin-n7-rank0-attempt/qualification.json`.

A separate marker-only constructor probe also times out: 110 s compute, 116.604 s total within its 150 s bound. Its first six-line chart takes 107.091 s inclusive, containing a 91.136 s determinant, 11.652 s inverse and 4.297 s product; the second determinant is active at termination. These nested observations diagnose that probe only. They are not summed, used as performance comparisons or retrospectively attributed to the earlier capture. The constructor remains incomplete, with no workspace, masters or arrays. Source: `validation/highpoint-thin-n7-phase-attribution/qualification.json`.

A private followup uses Symbolica's `RationalPolynomialField` for the matrix coefficient domain. All 22 selected exact tests pass; the rank-zero constructor completes in 58.442 s (66.660 s including guards, within the 150 s bound). This preparation diagnostic creates no workspace, masters or numerical arrays. It establishes no numerical or performance gate and is not applied to live production. Both earlier timeouts remain unchanged. Source: `validation/highpoint-thin-n7-rational-domain/qualification.json`.

29 Precision and high-point research

29.1 Typed direct E6: bounded common-mass fixtures

The unchanged private N2 direct genuine D6-pentagon engine is checked on common complex masses and zero invariants, with four- and six-component mixed Taylor shapes. Across ten groups, 65 lanes succeed and 65 intentionally reject; all 320 successful components satisfy the requested enclosure budget. Rejected lanes hide values. This extends the earlier one-axis prototype; it does not integrate the engine into the public reduction path.

Base type	Working / requested bits	Accepted / rejected lanes	Accepted components
f64	53 / 40	13 / 13	64
DoubleFloat	106 / 80	13 / 13	64
Float	128 / 80	13 / 13	64
Float	256 / 128	13 / 13	64
Float	384 / 128	13 / 13	64

Independent exact audit: 24,889 checks and 166 hashes, zero failures; the root independently reran that audit. Inputs distinguish 2^{-80} where the type permits it, retain mixed input precisions and verify scale-jet independence. Static Mixed4/Mixed6 and dynamic HyperDual metadata transport use Symbolica's Dualizer with NativeFloat workspaces; this is **not** `Workspace<StaticDual<T>>` qualification. Empty, odd, reordered, chunked, fresh, invalid-neighbor and independent concurrent workspaces pass compiled assertions; complete repeated arrays are not serialized.

Float256⟶384 arithmetic widths shrink for 102 positive coordinates and remain zero for 26. This comparison removes twice the retained truncation radius; it does not establish convergence of the fixed N2 mixed-cubic truncation. Instruction-lane visits are structural estimates. Unequal masses, nonzero invariants in this typed extension, generalized powers/dimensions, N4, contour/geometry certification and performance remain outside its scope. Production E6 failures remain unchanged. Source: `validation/e6-direct-n2-typed/`.

29.2 Exact child localization and mixed-jet counterexample

Two retained regular planar charts locate an ordinary child with Gram rank 2/determinant 0 and Cayley rank 4/determinant −4. Its multiplier vanishes at this exact ordinary input; generic symbolic aliasing still requests the singular PV child. This is preparation/identity evidence only: it does not identify the first nonfinite runtime instruction or evaluate a scalar master.

A separate exact mixed-jet counterexample has zero primal and first components but nonzero mixed cofactors −1/−1 and weights −2/−1 in the two charts. **Primal zero is insufficient for pruning.** Any future mask must certify the complete requested cofactor jet while retaining planar and chart regularity checks. No pruning implementation, successful N6 array, production change or performance result is claimed by these sources: `validation/highpoint-child-localization/`.

30 Independent pentagon adjudication

These checks establish selected mathematical values independently of an original-reference stability flag. They do not qualify every pentagon tensor, physical graph, runtime domain or numerical precision.

30.1 Rank-six Euclidean tensor

For squared masses $(4, 9, 16, 25, 36)$, $\mu = 3$ and numerator $q_0^5 q_3$, direct simplex integration gives native finite coefficient $-0.0024238378935447123i$. Independent coordinate mappings agree within 4.34×10^{-19}; reported quadrature error is at most 2.77×10^{-15}. The bound $F \geq 3.75$, angular integration and power counting exclude poles. This uses no reduction/master library; numerical error estimates are not interval certificates. See `validation/pv-high-rank/pentagon-simplex-oracle.json`. Historical binary64 errors remain 2.23×10^{-10} for TIRust and 3.40×10^{-10} for IREGI. Public rescue flags the primary unstable and verifies all 840 slots at 128/256 bits plus this coefficient below 10^{-13}. `validation/rescue-pentagon-rank6.json` retains the binary and assertions; full arrays were not printed and its exact build-source snapshot was not saved.

The vendored Symbolica baseline passes four scope tests, 57 core tests and all 1320 rank-seven slots at 128/256 bits, including the independent coefficient. Its source predates later precision/routing edits; its workspace compilation preserves 329 source/manifest hashes. The 109.44 s cold run, earlier 240 s timeout and 115.35 s isolated run are historical diagnostics. See `validation/symbolica-production-qualification.json`.

30.2 Physical canonical moment 54

For $gg \to HHH$, rank-four moment $q_0 q_3^3$ has UV degree -2; its UV pole must vanish. Original IBP instead returns approximately $3.33691766067 \times 10^{-6} i$ in native normalization. The 128/256-bit PV evaluation suppresses its OneLoop-normalized UV residual from 7.9×10^{-31} to 2.3×10^{-69}, with both IR slots zero. Its converged OneLoop finite coefficient is approximately $3.3474440772594743330 \times 10^{-13} + 2.9254797223941252086 \times 10^{-13} i$. This supports original PV/hybrid and rejects the original IBP result for this component. It is not a full physical tensor comparison. Raw precision evidence is in `validation/pv-high-rank/canonical-pentagon-precision.json canonical-pentagon-adjudication.json` records fixed-gate comparisons.

30.3 Standalone six-dimensional scalar

Use $J_1 = \frac{I_6}{i\pi^3}$, shifts $(0, e_0, e_1, e_2, e_3)$ and squared masses $(10, 12, 15, 18, 22)$. Here $F \geq \frac{39}{4}$, $\det R = 2600$ and $\det G = -16$.

Method	Finite J_1
Independent simplex	$-0.00017714013826400818060765213 10$
Original scalar/general APIs	-0.00014953025024128455; every mode reports stable

Increasing quadrature order and precision through 70 digits converges to the first value. Independent integration of ordinary D4 values over a common mass shift, with a separately integrated tail, agrees within 4.93×10^{-15}. A box-only half-line implementation also agrees to approximately 8.42×10^{-17} or better in the recorded checks. Ordinary D4 and coalesced D4/D6 normalization checks pass. Evidence and reproduction source are in `validation/shifted-pentagon/evidence.json`. Thirteen consolidated D6 tests cover eight genuine-master checks and five endpoints. Retained public f64 fixtures pass finite values, poles, scaling, rejected endpoints and lane isolation; the earlier quadrature-budget failure remains historical. Pre-vendor public DoubleFloat and Float128 tests now pass three scaled lanes at requested 80-bit accuracy and 10^{-23} normalized oracle tolerance. First common-mass Dual primal/derivative checks pass at 2×10^{-14}, with zero poles. DD/Float validation takes 165.69 s, exceeding its historical 150 s bound; this is not a benchmark. Build scope is in `validation/shifted-pentagon/public-core-qualification.json`. Paired vendored DD E6 passes exact allocation parity. Requested128 fails for the small-routing fixture; higher/mixed Dual, broader endpoints and native E6 remain unqualified. The original complex-box discrepancy is retained in `validation/shifted-pentagon/contours.json`.

A private input-enclosure extension passes five D0 pinches and two contractions at the single exact contour point $x = \frac{2}{5}$, Float256/request128: 40 rectangles and 20 rounded mass slots checked. Exact unshifted geometry remains assumed; no quadrature certificate or production E6 qualification follows.

31 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

31.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order (p^0,p^1,p^2,p^3). Propagators are $D_j=\left(q+p_j\right)^2-m_j^2+i0$, raised to positive integer powers. Generalized integrals use $d=4+\mathrm{idim}-2\varepsilon$ and the native raw measure; dimension shifts introduce powers of π. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes μ^2 to QCDLoop and μ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2)\le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

31.2 Native poles and MadLoop conversion

The native result order is

$$(U,I,J,F)=(\varepsilon_{\mathrm{UV}}^{-1},\varepsilon_{\mathrm{IR}}^{-1},\varepsilon_{\mathrm{IR}}^{-2},\mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite},\varepsilon^{-1},\varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank r to MadLoop order (finite, simple pole, double pole). Defining $a=\gamma_E+\ln\pi$, $b=\frac{a^2}{2}+\frac{\pi^2}{12}$ and $c_r=\frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}},S_{\mathrm{ML}},D_{\mathrm{ML}})=c_r(F+a(U+I)+bJ,U+I+aJ,J).$$

For a OneLOop-normalized scalar master (L_0,L_1,L_2), the scalar wrappers use

$$J=i\pi^2L_2,\quad U+I=i\pi^2(L_1-aL_2),$$

$$F=i\pi^2\left(L_0-aL_1+\left(\frac{a^2}{2}-\frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U,I,J,F)=(i\pi^2,-i\pi^2,0,0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

31.3 Polynomial and tensor interfaces

A rank bound R has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of q^0, q^1, q^2 in descending order; the exponent of q^3 is the remaining degree. Thus the start is $\left(1,q^0,q^1,q^2,q^3,\left(q^0\right)^2,q^0q^1,q^0q^2,q^0q^3,\left(q^1\right)^2,\ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

32 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

32.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.
- **Laurent coefficients:** IBP/PV factors and native normalization use cached Symbolica `series` templates and `coefficient` extraction. The three reuse checks are the pinned Symbolica `src/atom/core.rs` and `src/poly/series.rs` APIs; its `src/derivative.rs` pole-series tests; and existing use in `vakint/src/projection.rs` and OneLoopMaster's `src/sheet_exact.rs`. No separate Laurent convolution engine is needed. Symbolica also evaluates the independent Gamma/series triangle oracle.

32.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep D symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

32.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

Primary references: Alwall et al., appendix C.1 HEPKit TensorReducer Goode et al., tensor projectors Collins–Matsumoto, Weingarten recurrence NinjaRS report style. Local source audit: FeynKit's `feynkit-tensor/src` projector modules and `feynkit-graph/src/integrals.rs` original IREGI's `IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90` and `funlib.f90`.

33 Final private numerical checks and handoff

The following results extend the saved evidence. They do not apply the private seven-point implementation or scheduler patch to the public default reducer, and they introduce no new kernel timings. Earlier failed attempts, reference instabilities and inconclusive oracle or timing rows remain unchanged.

33.1 Seven-point scalar: one fixed ordinary lane

The private rational-polynomial implementation completes generic symbolic preparation independently at Float128 and Float256, with requested accuracy 53. It packs all 21 five-point children into one batch and returns all four native Laurent slots. Every slot agrees with the retained stable original IREGI IBP and Hybrid results at the unchanged dimensionally normalized 10^{-10} gate. The original PV result remains unstable and is not counted as agreement.

Saved numerical gate	Complete checks	Maximum normalized error
Original stable IBP / Hybrid, both precisions	16 Laurent slots	4.783×10^{-12}
Float128 / Float256 convergence	4 Laurent slots	9.102×10^{-34}

All three poles are exactly zero at both precisions. An independent audit reconstructs acceptance using exact rational squared inequalities, without importing the producer comparator. The full complex norm is compared: the tiny real finite component has tracked precision 2, so these results do not claim 128/256-bit componentwise accuracy. The previous fixed scalar oracle remains inconclusive. This is one private rank-zero case, not a qualification of N7 higher ranks, derivatives, generalized interfaces, numerical types, batches, physical graphs or performance.

Capture totals including final provenance checks are 153.623 s and 164.176 s, both within their predeclared 300 s bounds. They include generic preparation, workspace construction and diagnostics and are not benchmark measurements. Source: `validation/highpoint-thin-n7-rank0-rational-numeric/qualification.json`.

33.2 Uniform-branch scheduler: complete retained dependent corpus

Separate matching Symbolica/OneLoop/masters/core chains pass 60 requests, 30 per process. Exact complete records agree across schedulers and repeated or retained workspaces: eight ordinary N2/N3 rank0/1 batch1/32 families cover 1,584 tensor and 1,782 master complex values. Direct master controls cover 67 cases and 133 rows, including 106 successes and 27 explicit error/null rows. All six master outputs, requested Taylor components, represented values, precision, signed zero, diagnostics, masks and execution counters are checked. Independent saved-data and byte-copy audits pass; earlier compiler failures and the stricter admission-schema revision are retained.

This qualifies the frozen opt0 corpus only. The patch is not applied to production, and neither optimized performance nor full-domain acceptance is established. Fresh matched optimized chains are still needed: the older optimized closure predates a current Symbolica polynomial guard. Source: `validation/uniform-branch-dependent/qualification.json`.

34 Resumed native-master validation

Development resumed on 25 October 2026. The seven previously unrun constructor cases pass in the frozen default/selective builds, with two complete test listings and independent audits. All 646 bound inputs remained unchanged and all nine owned process groups exited. Together with twenty earlier ordinary cases, this completes the 27-case focused inventory across two separate runs. The earlier twenty cases were not rerun.

New constructor gate	Complete cases	Interpretation
Default / selective policy composition	3 + 4 passed	Construction and policy routing; no six-point integral values

The bounded run used 282.090 s including provenance and cleanup, within its unchanged 300 s limit. This is a diagnostic test duration, not a kernel benchmark. Source: `validation/native-f64-sixpoint-constructors/qualification.json`.

34.1 Complete native-policy tensor comparisons

Distinct baseline and selective-policy cores pass the original 38-request corpus: 30 TIRust requests and eight original IREGI requests. The eight ordinary bubble/triangle rank-zero/one families include singleton and batch-32 inputs. All complete tensor and captured master arrays are checked before accepting the policy; repeated workspaces and retained controls preserve full records.

Whole-call comparison	Complex slots / variant	Maximum normalized error
Complete tensor arrays	1,584	3.378×10^{-13} versus IREGI
Captured scalar masters	1,782	1.695×10^{-14}
Baseline versus candidate tensors	1,584	2.061×10^{-13}

The unchanged gate is 10^{-10}. D0, bubble derivatives, Dual and precision controls remain exact by represented values, precision and sign. This is an unoptimized frozen dependency context, not a current-Cargo or performance measurement. Source: `validation/native-f64-sixpoint-whole/qualification.json`.

34.2 Retained six-point behavior under both feature configurations

All ten numerical producers and eight saved gates pass in the default and selective combined source chains. The explicitly supported six-point parent and children use Symbolica. Checks include 24 ordinary paired cases, 4,400 complex Taylor slots, 560 rank-three original-reference comparisons and 280 128/256-bit convergence comparisons. The largest normalized errors are 2.426×10^{-12} against IREGI and 1.474×10^{-34} across precision. All 420 checked pole slots are exactly zero. Twenty-two lifecycle calls cover reordered/repeated/chunked samples, failures, empty calls and contractions; values, precision and sign match retained singleton results exactly.

Independent numerical and provenance audits pass; all 17 owned process groups exited. No f64/DoubleFloat high-point accuracy or public six-point Dual claim follows. Source: `validation/native-f64-sixpoint-regression/qualification.json`.

35 High-point numeric pilot and open qualifications

The private rank-zero six-point pilot evaluates one massive real physical point. Repeated and batched profiles do not increase the number of independent physical points. Missing arrays remain failures.

Numeric domain	Completed Laurent slots	Outcome at unchanged bounds
f64	0 / 40	All ten intended positive lanes rejected by cancellation guard; two invalid controls rejected.
DoubleFloat	40 / 40	Ten successful lanes, two invalid controls and one empty call pass.
Float128 dynamic Dual	24 / 240	One affine singleton passes; later schedule times out. Full domain fails.
Float256 dynamic Dual	0 / 240	Work deadline reached before a complete output. Precision-pair gate has zero comparisons.

DoubleFloat agrees with original IREGI within 8.175×10^{-14} and the retained Float256 scalar target within 6.420×10^{-30}. The available Float128 prefix has 24 complete complex coefficients, 18 exactly zero poles, and maximum dilation-identity error 1.013×10^{-34}. Two formal variables describe one physical dilation direction; the nonlinear mixed profile remains incomplete. Independent checks reconstruct all 64 available outputs and 1,736 exact input components, including seeds below binary64 resolution.

Each domain retained its 180-second aggregate bound and 20-second postwork reserve; no failed attempt was retried or extended in this pilot. Its timings are lifecycle diagnostics. Public high-point Dual accuracy, derivative error transport, general independent derivative directions and full convergence remain unqualified. Source: `validation/highpoint-six-point-numeric-types/qualification.json`.

35.1 Exact routing preparation

A private optimization delegates bounded affine routing matrices to Symbolica's rational-polynomial matrix solver. All 16 selected exact algebra tests pass, including actual generic N6/N7 scalar-compiler routing, generalized power/dimension identities, Gaussian coefficients and explicit fallback for opaque or nonpolynomial entries. The initial missing-environment build failure and interrupted scheduling attempt remain recorded. This is not a full-core numerical or performance qualification and is not applied to production. Source: `validation/routing-rational-preparation/qualification.json`.

The optional native policy passes the current locked-Cargo checks detailed below. The restricted public seven-point proposal and shared exact-master preparation cache remain separate candidates. Matched optimized scheduler products pass fresh numerical admission. Their timing cohort is retained below, with final performance eligibility false because of substantial sibling-core activity. Existing IREGI/COLLIER timings retain their contexts and TIRust-numerator ratios. The full goal remains active: complete N6/N7 rank/power/dimension coverage, high-point precision/Dual qualification, broader physical/exceptional checks and performance parity per family remain required.

36 Optional native policy: current Cargo integration

The Rust working tree integrates the explicit `MasterExecutionPolicy::NativeF64Abc` policy through the `native-f64-abc` feature. Oneloopmaster's generated Rust evaluators handle ordinary f64 A0/B0/C0 requests instruction-major across active lanes. D0, bubble derivatives, DoubleFloat, arbitrary precision and every static or dynamic Dual layout retain Symbolica under the selective policy. Selecting the policy without its feature returns an explicit preparation error.

Actual locked, offline Cargo builds and runtime checks pass for the default and selective configurations, without private capture instrumentation or substituted project archives. Current-source metadata records verify the expected feature forwarding. The executable copies are taken from their exact Cargo artifact records before another configuration can overwrite a shared example path.

Current configuration	Selected exact tests	Public five-lane triangle probe
Default	9 / 9 pass	Default and explicit Symbolica records agree exactly.
Selective feature	17 / 17 pass	Default, Symbolica and native policies pass complete tensor checks.

These 26 selected cases cover policy routing, ordered Dual layouts, independent scratch, precision integrity and family exclusions. They are a focused subset, not a claim that every project test was rerun. The N3/rank-one probe uses three retained physical points, one invalid-scale neighbor and one repeated point. All 400 complex Laurent comparisons against original IREGI pass the unchanged dimensionally normalized 10^{-10} gate, with maximum error 2.346×10^{-13}. The 80 direct selected-versus-Symbolica comparisons have maximum error 1.486×10^{-13}. Invalid lanes contain no successful tensor array; values, precision, signs and execution counters remain checked.

36.1 Retained compatibility failure

Legacy feature configuration	Metadata outcome	Qualification boundary
Broad `native-batch`	Timeout	Reached the fixed 240 s frontend limit; group cleanup and unchanged sources verified.
Broad + selective	Unrun	Later matrix step was not attempted after the timeout.

The failed matrix attempt remains preserved, with no retry or extended bound. A separate reviewed continuation admits the already-passed default/selective metadata rows for their actual builds and numerical checks. Neither the complete legacy matrix nor broad native runtime is qualified by this checkpoint. The large legacy D0 evaluator received no LLVM code-generation request.

All 39 owned process groups exited. Independent source, artifact, numerical and saved-data audits pass for the stated native-only scope. Six-point numerical regressions remain the separate preceding capsule; no new N6/N7 domain or performance claim follows from the Cargo checks. Backend selection is exposed through the Rust API; CLI/Python selection is future work. Default constructors retain their backend behavior. The explicit six-point parent and children use Symbolica. Source and reproduction commands: `validation/native-f64-sixpoint-current-cargo/qualification.json` and `docs/native-masters.md`.

37 Fresh scheduler experiment: retained, ineligible

Final performance eligibility: false. The experiment ran on CPU 34 while its SMT sibling, CPU 290, had substantial unrelated activity (69.6% non-idle over the enclosing observation interval). All mathematical checks pass; timing arithmetic and raw gates are retained. These measurements establish no uncontended parity or production-default improvement.

This private comparison pins `2c1f334` and generic Symbolica masters in the same inherited optimized dependency context for both TIRust variants. It does not measure the later opt-in generated f64 policy. Fourteen matched compiler products, captured and uncaptured complete-array admission, and six-process preflight pass. The earlier linker-map admission failure remains recorded; its five valid products were reused and nine were built subsequently.

The one-shot run retains all eight massive Euclidean bubble/triangle families, 48 balanced quartets per family, 1,768 requests and 1,600 timed replies. Transport lies outside the timed kernel. All six processes exit successfully and final controls pass. No short or noisy family is removed or rerun.

37.1 Absolute kernel times

Medians in **µs per complete batch call**. For batch 32, divide by 32 for the amortized per-sample time; the machine-readable record contains both units. These absolute medians and the paired ratio medians on the next page are different statistics: ratios are not calculated by dividing this table.

N / rank / batch	TIRust baseline	TIRust candidate	IREGI	COLLIER
2 / 0 / 1	2.209	2.278	0.944	0.760
2 / 0 / 32	43.881	40.940	28.438	24.661
2 / 1 / 1	7.099	7.291	3.188	1.199
2 / 1 / 32	264.385	259.023	157.852	65.059
3 / 0 / 1	31.145	31.746	7.818	6.235
3 / 0 / 32	412.617	382.028	164.619	130.904
3 / 1 / 1	43.299	42.683	16.052	9.677
3 / 1 / 32	964.618	906.794	525.433	314.702

The raw candidate/baseline observations include about 6% lower time for N2/rank0/batch32 and N3/rank1/batch32, with 2–3% singleton regressions for N2/rank0 and N3/rank0. This mixed result does not support globally adopting the scheduler candidate, even before considering contention. The production scheduler remains unchanged. Higher-precision and Dual performance are not measured in this cohort.

Reproduce the saved-evidence checks without evaluating new integrals: `python3 validation/scheduler-four-engine-fresh-2c1/verify.py`. Then run `python3 validation/scheduler-four-engine-fresh-2c1-supplement-check.py` to check the two historical nested manifests omitted by the original seal. Machine-readable source: `validation/scheduler-four-engine-fresh-2c1/performance.json`.

38 Fresh scheduler experiment: all paired ratios

Every numerator is TIRust. **B** denotes baseline and **C** the private candidate. Each large cell gives the paired median and 99% bootstrap interval. Color shows the raw ratio only; **every cell is ineligible for final acceptance** because of the shared-core context. Short blocks remain amber.

N / rank / batch	B / IREGI	C / IREGI	B / COLLIER	C / COLLIER	C / B
2 / 0 / 1	2.334 2.315–2.340 Slower	2.412 2.388–2.420 Slower	2.894 2.879–2.919 Slower	2.993 2.973–3.013 Slower	1.033 1.030–1.036 Slower
2 / 0 / 32	1.531 1.497–1.542 Noise	1.435 1.400–1.446 Noise	1.756 1.727–1.786 Noise	1.648 1.622–1.668 Noise	0.935 0.930–0.945 Lower time
2 / 1 / 1	2.313 2.267–2.345 Noise	2.336 2.311–2.350 Slower	5.850 5.800–5.908 Slower	5.936 5.772–6.012 Noise	1.010 0.995–1.022 Noise
2 / 1 / 32	1.671 1.668–1.677 Short block	1.638 1.635–1.644 Short block	4.061 4.051–4.068 Short block	3.979 3.974–3.986 Short block	0.981 0.980–0.984 Short block
3 / 0 / 1	3.982 3.967–3.995 Slower	4.062 4.052–4.072 Slower	4.991 4.966–5.000 Slower	5.085 5.057–5.102 Slower	1.019 1.016–1.021 Slower
3 / 0 / 32	2.433 2.415–2.478 Short block	2.201 2.180–2.294 Short block	3.140 3.112–3.159 Short block	2.830 2.819–2.875 Short block	0.905 0.901–0.923 Short block
3 / 1 / 1	2.719 2.630–2.735 Noise	2.641 2.572–2.678 Noise	4.518 4.435–4.561 Noise	4.391 4.305–4.454 Noise	0.977 0.961–0.983 Noise
3 / 1 / 32	1.818 1.746–1.837 Noise	1.717 1.648–1.725 Noise	3.059 2.818–3.069 Noise	2.876 2.668–2.883 Noise	0.939 0.937–0.941 Lower time

Raw gates: **12 regressions, 16 noise-inconclusive, 10 short-block-inconclusive and 2 improvements**. The short families are N2/rank1/batch32 (one short block) and N3/rank0/batch32 (50 short blocks). The unchanged minimum block duration is 100 ms. All 384 quartets and all 40 confidence intervals remain available, including these failures of timing admission.

Independent saved-data review reproduces the numerical admissions, complete request accounting, quartet orders and all ratio arithmetic. A separate environment review sets final acceptance false without rewriting raw gates. A future cohort needs fresh placement checks and a new measurement protocol; no extrapolation to current native-policy or high-point performance is made.

Source and full scope: `validation/scheduler-four-engine-fresh-2c1/qualification.json`.